

Absolute Purity for Motivic Cohomology in Dimensions at Most Three

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Abstract

We prove absolute purity for the intrinsic motivic coefficient $H = H\mathbf{Z}^{\mathbf{A}^1} = s_0 KGL$. Integral point purity holds for regular noetherian local rings with fraction fields of characteristic zero in dimension at most two, and in dimension at most three under excellence, with the corresponding flat regular-pair statements. For an invertible prime ℓ , purity modulo every ℓ^m holds in local or ambient dimension at most $\ell + 2$. In every finite dimension, integral purity also holds for local algebras regular over a mixed-characteristic DVR and for flat regular $\mathbf{Z}_{(p)}$ -pairs with regular special fibres. The proofs combine filtered Gysin maps and Adams operations, valuative comparison and descent, and detection by smooth function fields.

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1 Main results and conventions

Let

$$H_T = H\mathbf{Z}_T^{\mathbb{A}^1} = s_0 KGL_T.$$

This is the Cartesian oriented absolute coefficient of (Bachmann et al., 2025, arXiv v1, Definition 4.15, Theorem 9.3(1),(2), and Corollary 9.7). For a regular closed immersion $i : Z \hookrightarrow X$ with normal bundle N_i , we study the fundamental-class transformation

$$\mathrm{pur}_i(H_X) : \mathrm{Th}(-N_i) \otimes H_Z \longrightarrow i^! H_X. \quad (1)$$

For constant codimension c , orientation identifies its source with $H_Z(-c)[-2c]$. The local and ambient dimensions in the following statements are Krull dimensions.

All derived complexes are cohomologically indexed: $H^j(C[a]) = H^{j+a}(C)$, and their associated spectra satisfy $\pi_i = H^{-i}$. Put $T = \Sigma^{2,1}\mathbf{1}$ and write

$$\begin{aligned} A(q)(Y) &= \mathbf{Z}(q)^{\mathbb{A}^1}(Y), & A_n(q) &= A(q) \otimes^{\mathbf{L}} \mathbf{Z}/n, \\ A(q)(Y)[2q] &= \mathrm{map}(\mathbf{1}_Y, T^q \otimes H_Y). \end{aligned}$$

Negative-weight values are zero. For a presheaf P and a closed subset $Z \subset Y$ with quasi-compact complement, set

$$P_Z(Y) = \mathrm{fib}(P(Y) \longrightarrow P(Y \setminus Z)).$$

Finite coefficients commute with these supported fibres. For a general $E \in \mathbf{SH}(Y)$, $E(U)(q)$ denotes its mapping spectrum on a smooth test U , with the Tate twist alone: $E(U)(q) = \mathrm{map}(\Sigma^\infty U_+, E(q))$. Thus $T^q E = E(q)[2q]$, and no cohomological shift $[2q]$ is included in $E(U)(q)$. Continuous local test schemes are affine inverse limits of smooth test schemes; their values are the corresponding filtered colimits. Only the coefficient-module values are written as complexes in $D(\mathbf{Z})$. All schemes evaluated by these presheaves are qcqs.

For a regular local ring R , the condition $\mathrm{char} \mathrm{Frac}(R) = 0$ is equivalent to $\mathrm{char} R = 0$: the ring is a domain (The Stacks Project Authors, 2026, Tag 00NP) and embeds into its fraction field. The residue field is allowed to have positive characteristic.

Theorem A (Integral purity). *Let R be a regular noetherian local ring of dimension d , with closed point s and residue field κ , and assume that $\mathrm{char} \mathrm{Frac}(R) = 0$. If $d \leq 2$, or if $d \leq 3$ and R is excellent, then*

$$H_\kappa(-d)[-2d] \xrightarrow{\sim} s^! H_R \quad \text{in } \mathbf{SH}(\kappa).$$

In particular, after choosing regular parameters, its value in every weight q is the equivalence

$$A(q-d)(\kappa)[-2d] \xrightarrow{\sim} A(q)_s(R).$$

For a regular closed immersion $i : Z \hookrightarrow X$ between regular noetherian schemes flat over $\mathbf{Z}$, $\mathrm{pur}_i(H_X)$ is an equivalence if $\dim X \leq 2$, or if $\dim X \leq 3$ and both schemes are excellent.

Theorem B (Prime-power purity). *Let R be a regular noetherian local ring of dimension d , with closed point s and residue field κ , and let ℓ be a prime invertible in R . If $d \leq \ell + 2$, then for every $m \geq 1$ the point-purity transformation for H_R/ℓ^m is an equivalence. Thus, for every weight q ,*

$$A_{\ell^m}(q-d)(\kappa)[-2d] \xrightarrow{\sim} A_{\ell^m}(q)_s(R).$$

For a regular closed immersion $i : Z \hookrightarrow X$ between regular noetherian schemes, if ℓ is invertible on X and $\dim X \leq \ell + 2$, then

$$\mathrm{Th}(-N_i) \otimes (H_Z/\ell^m) \xrightarrow{\sim} i^!(H_X/\ell^m) \quad (m \geq 1).$$

These assertions impose no excellence, flatness or residue-field perfection hypothesis.

Theorem C (Purity over a regular DVR map). *Let R be a regular noetherian local ring of finite dimension d , with closed point s and residue field κ . Suppose there is a local regular ring map $V \rightarrow R$ from a mixed-characteristic DVR of residue characteristic p . Then*

$$H_\kappa(-d)[-2d] \xrightarrow{\sim} s^! H_R.$$

Its integral value in every weight is $A(q-d)(\kappa)[-2d] \simeq A(q)_s(R)$, and

$$A_{p^m}(q)_s(R) \simeq W_m \Omega_{\kappa, \log}^{q-d}[-(q+d)] \quad (m \geq 1),$$

with zero value for $q < d$. In particular this applies to every unramified regular local ring of mixed characteristic $(0, p)$, that is, with $p \notin \mathfrak{m}_R^2$.

For a regular closed immersion $i : Z \hookrightarrow X$ between regular noetherian $\mathbf{Z}_{(p)}$ -schemes of finite Krull dimension, assume that both schemes are flat over $\mathbf{Z}_{(p)}$ and that both special fibres are regular. Then $\mathrm{pur}_i(H_X)$ is an integral equivalence. No dimension bound or excellence assumption is imposed.

Here a regular ring map is flat with geometrically regular fibres. In Theorem A, positive residue characteristic p is nonzero in R ; flatness supplies this condition for its pair statement. Theorem C also allows DVR bases ramified over $\mathbf{Z}_{(p)}$.

Gabber's étale absolute purity theorem has no dimension bound for invertible finite coefficients (Riou, 2014, Théorème 3.1.1). The motivic comparison with étale cohomology controls only a truncation. We use it to bound the Gysin defect, then use the filtered K -theory Gysin map and exact Adams eigenweights to eliminate that defect. At the residue prime, polynomial invariance on F -smooth valuation rings supplies comparison with the $\mathbb{A}^1$ -invariant theory. Regular surfaces admit a quadratic-transform argument without excellence; in dimension three, resolution supplies the bounded valuation models needed for two distinct descent arguments. Detection by smooth function fields then passes from values to objects and, by composition, to regular pairs. With $F_p(q) = \mathbf{Z}/p(q)^{\mathrm{cdh}}$, Theorem 4.4 also gives the independent comparison $F_p(q)(Y) \simeq A_p(q)(Y)$ for every excellent noetherian scheme Y of dimension at most three, every prime p , and every $q \geq 0$. It allows singularities, arbitrary residue fields and arbitrary ramification; its dimension-three proof invokes Cossart–Piltant's resolution theorem. Theorem 4.2 supplies the regular-surface comparison without excellence. The endpoint Adams argument and these residue-characteristic comparisons are the main mechanisms developed below. Levine proves integral cycle-complex localization for finite-type schemes over regular noetherian one-dimensional bases (Levine, 2001, preprint, Theorem 0.7 and Section 6). Bouis and Kundu prove a $\mathbf{Z}/p^m$ generic-fibre equivalence for ind-smooth schemes over perfectoid valuation rings of residue characteristic p (Bouis and Kundu, 2025, arXiv v2, Corollary 5.9 and Remark 5.10) in their non- $\mathbb{A}^1$ -invariant motivic theory, distinct from H . Fundamental-class DVR purity for Spitzweck's coefficient is proved in the mixed-characteristic DVR setup of Frankland–Spitzweck's Section 4 (Frankland and Spitzweck, 2026, arXiv v4, Section 4, Proposition 8.3 and Corollary 8.4). For every mixed-characteristic DVR, the internal dimension-one case of Theorem A supplies this input; Section 6 extends it to regular DVR maps by Popescu approximation and continuity.

We use the six operations, the exactness and naturality of purity in its coefficient, compatibility with twists, composition, transverse base change and smooth exceptional base change of (Déglise et al., 2021b, arXiv v3, Paragraphs 2.1.9–2.1.10 and 4.3.1, Proposition 2.5.4(i),(ii)(a), Theorem 3.3.2(iii), Proposition 4.3.10(i), and Remark 4.3.8(i)). For continuous tests these compatibilities are applied on finite-type models and passed to the limit (Hoyois, 2015, arXiv v5, Lemma A.7(1)).

Comparison with Spitzweck’s coefficient. Let $a_X : X \rightarrow \mathrm{Spec} \mathbf{Z}$ and put $M_X^{\mathrm{Spi}} = a_X^* M_{\mathbf{Z}}^{\mathrm{Spi}}$. The integral comparison over $\mathbf{Z}$ follows from Bachmann’s effective-cover theorem (Bachmann, 2025, arXiv v2, Theorem 1.1(1), Example 1.3 and Lemma 2.1), as recorded in (Bachmann et al., 2025, arXiv v1, Remark 4.14). Together with its Theorem 9.3(2), this gives

$$M_X^{\mathrm{Spi}} \xrightarrow{\sim} a_X^* s_0 KGL_{\mathbf{Z}} \xrightarrow{\sim} s_0 KGL_X = H_X$$

for every qcqs scheme X , including all local tests used below. Spitzweck’s construction directly over a mixed-characteristic Dedekind domain agrees with this absolute pullback by (Spitzweck, 2018, Theorem 8.25, p. 88; arXiv v3, Theorem 9.19). These are integral comparisons of coefficient objects; finite reductions transport them by exactness.

For any coefficient morphism $u_X : M_X \rightarrow H_X$, naturality of the intrinsic fundamental-class transformation gives the square

$$\begin{array}{ccc} \mathrm{Th}(-N_i) \otimes i^* M_X & \xrightarrow{\mathrm{pur}_i(M_X)} & i^! M_X \\ \downarrow \mathrm{id} \otimes i^* u_X & & \downarrow i^! u_X \\ \mathrm{Th}(-N_i) \otimes i^* H_X & \xrightarrow{\mathrm{pur}_i(H_X)} & i^! H_X. \end{array}$$

This is (Déglise et al., 2021b, arXiv v3, Paragraphs 2.5.1 and 4.3.1). If u_X is an equivalence, both vertical arrows are equivalences, so intrinsic purity transports without an independent purity-compatibility hypothesis. Equality of particular oriented Tate formulas additionally requires alignment of the chosen Thom isomorphisms. An E_∞ comparison is needed to identify coefficient algebras or their module categories; no such upgrade of the cited object comparison is asserted here. Other independently constructed coefficients require their own integral recognition on the bases in question, compatible with their Cartesian identifications. Rational or prime-to-characteristic comparison alone does not supply it.

2 Filtered purity and invertible prime powers

The absolute KGL spectrum has effective covers f_j , slices s_j and Bott identifications

$$s_q KGL_Y = T^q \otimes H_Y, \quad T \otimes f_j KGL_Y \simeq f_{j+1} KGL_Y, \quad g^* f_j KGL_Y \simeq f_j KGL_{Y'}.$$

Its values form the natural filtration

$$\begin{aligned} \mathrm{Fil}^j KH(Y) &= \mathrm{map}(\mathbf{1}_Y, f_j KGL_Y), \\ \mathrm{Fil}^j KH(Y) &= KH(Y) \quad (j \leq 0), \\ \mathrm{gr}^q KH(Y) &= A(q)(Y)[2q]. \end{aligned}$$

The supported filtration is $\mathrm{Fil}^j KH_Z(Y) = \mathrm{fib}(\mathrm{Fil}^j KH(Y) \rightarrow \mathrm{Fil}^j KH(Y \setminus Z))$. These assertions are (Bachmann et al., 2025, arXiv v1, Section 2.1, (4.7), Remark 4.20, Theorem 4.39(1), and Theorem 9.3(1)). The filtration is finitary, and $\mathrm{Fil}^j KH(Y)$ is $(j - a)$ -connective when $\dim_v Y \leq a$ (Bachmann et al., 2025, arXiv v1, Theorem 1.1(1),(3) and

Corollary 9.5). For locally noetherian schemes, valuative and Krull dimensions agree (Elmanto et al., 2021, arXiv v3, Proposition 2.3.2(9)). The graded-piece connectivity therefore gives

$$A(q)(Y) \in D^{\leq q+a} \quad (\dim_v Y \leq a). \quad (2)$$

Indeed $\mathrm{gr}^q KH(Y)$ is $(q - a)$ -connective. Reduction by finite coefficients preserves this upper bound.

Fix a prime ℓ and, on $\mathbf{Z}[1/\ell]$ -schemes, put

$$G_\ell(q)(Y) = R\Gamma_{\text{ét}}(Y, \mu_\ell^{\otimes q}), \quad T_\ell(q) = L_{\text{cdh}}\tau^{>q}G_\ell(q).$$

The multiplicative comparison is

$$A_\ell(q) = L_{\text{cdh}}\tau^{\leq q}G_\ell(q), \quad b : A_\ell(q) \longrightarrow G_\ell(q)$$

by (Bachmann et al., 2025, arXiv v1, Theorem 1.1(4) and Proposition 6.8). We use the arc-descent assertion for étale cohomology as applied in (Bachmann et al., 2025, arXiv v1, proof of Proposition 5.20), which invokes (Bhatt and Mathew, 2021, Theorem 5.4). It implies cdh descent. Consequently $A_\ell(q) \rightarrow G_\ell(q) \rightarrow T_\ell(q)$ is a cofiber sequence. Limits and filtered colimits preserve the lower bound, so

$$T_\ell(q)(Y) \in D^{\geq q+1}. \quad (3)$$

Every cdh covering sieve of a field is maximal. For every field L with ℓ invertible,

$$A_\ell(q)(L) = \tau^{\leq q}G_\ell(q)(L). \quad (4)$$

We use Gabber–Riou étale purity with its fundamental class and arbitrary-base-change compatibility (Riou, 2014, Définitions 2.1.1 and 2.3.1, Proposition 2.2.3.1, and Théorème 3.1.1); excellence is not required.

Lemma 2.1 (Coefficient algebras and supported module actions). *For every qcqs Y , H_Y is an E_∞ algebra over $c(H\mathbf{Z})$, where $c : \mathrm{Sp} \rightarrow \mathbf{SH}(Y)$ is the constant symmetric monoidal functor. Hence for every prime ℓ , including 2,*

$$H_Y/\ell \simeq H_Y \otimes_{c(H\mathbf{Z})} c(H\mathbf{F}_\ell)$$

is an E_∞ ring. The analogous derived coefficient algebra exists over $c(H\mathbf{Z}/\ell^m)$ for every $m \geq 1$.

For a regular closed immersion $i : Z \rightarrow Y$ of qcqs schemes, pur_i is a morphism of i^ -bimodules. For a cartesian motivic E_∞ ring E it therefore lifts to the purity transformation on its module categories. In particular the purity maps for KGL , H , and the derived coefficient algebras H/n are linear over their pullbacks to Z .*

*Let Y be qcqs, E an E_∞ ring in $\mathbf{SH}(Y)$, and let $i : Z \rightarrow Y$, $j : U \rightarrow Y$ be complementary closed and open immersions. Then $i_*i^!E \rightarrow E \rightarrow j_*j^*E$ is a fiber sequence of E -modules. On values it is the fiber sequence of $E(Y)$ -modules*

$$E_Z(Y) \longrightarrow E(Y) \longrightarrow E(U).$$

*The i^*E -action on $i^!E$ for which purity is linear corresponds to the cup product with supports, namely the $E(Y)$ -module structure on $\mathrm{fib}(E(Y) \rightarrow E(U))$. For assertions on homotopy groups, a homotopy-commutative ring suffices.*

Proof. Construction 4.35 of Bachmann–Elmanto–Morrow supplies the E_∞ comparison $R\Gamma_{\text{Nis}}(-, \mathbf{Z}) \rightarrow \mathbf{Z}(0)^{\mathbb{A}^1}$ on smooth schemes over a regular noetherian base (Bachmann et al., 2025, Construction 4.35 and Remark 4.37). Over $\text{Spec } \mathbf{Z}$, evaluation gives an E_∞ map $H\mathbf{Z} \rightarrow \text{map}(\mathbf{1}, s_0 KGL_{\mathbf{Z}})$. The symmetric monoidal adjunction for c and its lax monoidal right adjoint lifts to algebras, giving $c(H\mathbf{Z}) \rightarrow H\mathbf{Z}$. The zero slice is the absolute E_∞ algebra specified by (Bachmann et al., 2025, Theorem 1.6(1),(2)); pulling back this map along $Y \rightarrow \text{Spec } \mathbf{Z}$ gives the assertion. Relative tensor with $c(H\mathbf{Z}/n)$ computes the cofiber of n on the $H\mathbf{Z}$ -module and remains an E_∞ algebra. In particular the argument uses no ring structure on $\mathbf{S}/2$. On represented values this is the derived coefficient structure of (Bachmann et al., 2025, Remark 4.38).

Linearity follows from (Frankland and Spitzweck, 2026, arXiv version 4, Lemma 8.1 and Corollary 8.2); the cartesian property of the spectra used here is part of the absolute construction in section 2 and the coefficient construction above.

The projection formula makes $i_* = i_!$ linear over $\mathbf{SH}(Y)$. The linear adjunction $i_* \dashv i^!$ lifts to the adjunction on modules over i^*E and E (Lurie, 2017, Corollary 7.3.2.7 and Example 7.3.2.8). The lax action on the right adjoint is adjoint to the projection-formula map formed with the counit; this is exactly the exchange map $i^!E \otimes i^*E \rightarrow i^!(E \otimes E)$ (Lurie, 2017, Remark 7.3.2.9) and (Déglise et al., 2021b, arXiv version 3, Paragraph 2.1.10). Thus $i^!E$ is an i^*E -module, its pushforward is an E -module, and the counit $i_*i^!E \rightarrow E$ is E -linear. The object j_*j^*E is an E -algebra through its unit. The composite of these two maps is null already in modules, since

$$\text{Map}_{\text{Mod}_E}(i_*i^!E, j_*j^*E) \simeq \text{Map}_{\text{Mod}_{j^*E}}(j^*i_*i^!E, j^*E)$$

and $j^*i_* = 0$. Hence the counit factors through the module fiber of $E \rightarrow j_*j^*E$. Its underlying factorization is the localization equivalence, so it is an equivalence in modules.

The functor $\text{map}(\mathbf{1}_Y, -)$ is exact and lax monoidal, and sends this sequence to $E(Y)$ -modules. Uniqueness of the module fiber identifies its action with the cup product with supports. The projection formula identifies the value of the action on $i^!E$ with this action on $i_*i^!E$. $\square$

Cup products of two supports are formal for Zariski sheaves of E_∞ rings: $P_Z \otimes P_W \rightarrow P_{Z \cap W}$. A multiplicative comparison preserves these products. We apply this to the graded coefficient values and to the comparison b .

Lemma 2.2 (Filtered Gysin maps and connectivity). *Let $i : Z \rightarrow Y$ be a regular closed immersion of codimension c between regular noetherian schemes of finite dimension, with normal bundle N . The maps*

$$\Gamma_j = \text{pur}_i(f_j KGL_Y) : \text{Th}(-N) \otimes i^*f_j KGL_Y \longrightarrow i^!f_j KGL_Y$$

form a map of $\mathbf{Z}$ -indexed filtered objects. With the Thom identification of the source, this gives

$$\text{Gys}^j : \text{Fil}^{j-c} KH(Z) \longrightarrow \text{Fil}^j KH_Z(Y).$$

In particular a specified trivialization of N gives the identification by T^{-c} and Bott periodicity. Without such an identification the first display is the intrinsic, Thom-twisted statement. Its graded pieces are

$$A(q-c)(Z)[-2c] \longrightarrow A(q)_Z(Y),$$

shifted by $[2q]$. The same assertions hold after every finite coefficient $\mathbf{S}/n$, without requiring n to be invertible on Y .

Let $Y = \operatorname{Spec} R$ for a regular noetherian local ring of dimension d , with residue field κ , and $Z = s$ its closed point. A regular system of parameters trivializes N . Then

$$\operatorname{pur}_s(KGL_Y) \circ \beta^{-d} = \pm \eta'_s,$$

where η'_s is the Cisinski–Déglise class; consequently Gys^j is an integral equivalence for $j \leq 0$, and remains an equivalence after every finite coefficient.

For $d \geq 1$, put integrally,

$$\Psi^j = \operatorname{fib}(\operatorname{Fil}^{j-d} KH(\kappa) \xrightarrow{\operatorname{Gys}^j} \operatorname{Fil}^j KH_s(Y)).$$

Then Ψ^j is $(j - d - 1)$ -connective, so Ψ^{q+1} is $(q - d)$ -connective and $\Sigma\Psi^{q+1}$ is $(q + 1 - d)$ -connective. The same bounds hold for the filtered fiber Ψ_n after every finite coefficient $\mathbf{S}/n$, without an invertibility hypothesis on n . Moreover $\Psi^j = 0$ and $\Psi_n^j = 0$ for $j \leq 0$.

Proof. Naturality of (1) along $f_{j+1} \rightarrow f_j$ makes Γ filtered. Base change of the covers is section 2. A trivialized $\operatorname{Th}(-N)$ is T^{-c} , so Bott periodicity identifies its tensor with $f_j KGL_Z$ with $f_{j-c} KGL_Z$. Localization identifies the value of $i^! f_j KGL_Y$ at Z with the supported filtration (Déglise et al., 2021b, arXiv version 3, Proposition 2.2.10). Taking cofibers in the filtration and using exactness of purity gives $\operatorname{pur}_i(s_q KGL_Y)$; the slice formula and compatibility with T^q give the stated graded Gysin maps. These are exact constructions, so finite coefficients commute with them.

The bottom equivalence. Both maps $KGL_\kappa \rightarrow i^! KGL_Y$ are KGL_κ -linear. Such a map is determined by the image of 1 in

$$\pi_0 \operatorname{map}(\mathbf{1}_\kappa, i^! KGL_Y) = K_0^s(Y) = K_0(\kappa) = \mathbf{Z}.$$

The Cisinski–Déglise class takes 1 to $[\mathcal{O}_s]$, and is an equivalence for a closed immersion of regular noetherian finite-dimensional schemes; on supported K -theory it is dévissage (Cisinski and Déglise, 2019, arXiv version 4, Definition 13.5.4, Proposition 13.6.1, and Theorem 13.6.3). For $t = (t_1, \dots, t_d)$, compare the graph $\Gamma_t : Y \rightarrow \mathbb{A}_Y^d$ with the zero section 0_Y . The pullback of 0_Y is s , and the square is Tor-independent because the Koszul complex of t resolves $\mathcal{O}_s$. Transverse base change gives $\eta_s = \Gamma_t^* \eta_{0_Y}$. Smooth-pair purity makes the class of 0_Y a generator of $K_0^{0_Y}(\mathbb{A}_Y^d) = K_0(Y) = \mathbf{Z}$; it is $\pm[\operatorname{Koszul}(T)]$. Pullback gives $\pm[\operatorname{Koszul}(t)] = \pm[\mathcal{O}_s]$, as required. For $j \leq 0$, both value filtrations are the unfiltered spectra. Finally $K = KH$ for regular noetherian rings (Weibel, 2013, Chapter IV, Corollary 12.3.1), so the value map is the dévissage equivalence up to sign.

Connectivity. Put $U = Y \setminus \{s\}$. Its dimension is $d - 1$: the upper bound follows by appending the closed point to a chain; the lower bound follows from a maximal prime chain in the regular local ring. The filtration on U is the value filtration of section 2, including when U is not affine. Thus $\operatorname{Fil}^j KH(Y)$ is $(j - d)$ -connective and $\operatorname{Fil}^j KH(U)$ is $(j - d + 1)$ -connective. Their fiber is $(j - d)$ -connective. The source $\operatorname{Fil}^{j-d} KH(\kappa)$ is also $(j - d)$ -connective. The fiber of a map between spectra with this connectivity is $(j - d - 1)$ -connective. Finite coefficient cofibers preserve all these bounds and commute with fibers. The bottom vanishing follows from the preceding dévissage equivalence. $\square$

For a regular noetherian local ring R of dimension d , write $X = \operatorname{Spec} R$, s for its closed point, κ for its residue field, and $U = X \setminus \{s\}$. Use regular parameters for the Thom identification and set

$$\Delta(q) = \operatorname{cofib}(A(q - d)(\kappa)[-2d] \xrightarrow{\operatorname{pur}_s} A(q)_s(R)), \quad \Delta_n(q) = \Delta(q)/n. \quad (5)$$

Linearity over $c(H\mathbf{Z})$ and exactness of purity make $\Delta(q)$ an object of $D(\mathbf{Z})$, and identify $\Delta_n(q)$ with the Gysin cofiber for $A_n(q)$. Exactness of the filtered construction gives

$$\mathrm{gr}^q \Psi_n = \Delta_n(q)[2q - 1].$$

Global filtered statements retain the Thom twist; the parameter identification is used only at regular local points.

Lemma 2.3 (Supported comparison and the étale fundamental class). *For every closed pair $Z \subset Y$ of qcqs $\mathbf{Z}[1/\ell]$ -schemes, the comparison $A_\ell(q)_Z(Y) \rightarrow G_\ell(q)_Z(Y)$ is an isomorphism on H^n for $n \leq q$, and injective for $n = q + 1$.*

Let R be regular noetherian local of dimension $d \geq 1$, with ℓ invertible. Then there is a unit $a \in (\mathbf{Z}/\ell)^\times$, independent of q , such that on every H^n the composite

$$A_\ell(q - d)(\kappa)[-2d] \xrightarrow{\mathrm{pur}_s(q)} A_\ell(q)_s(X) \xrightarrow{b} G_\ell(q)_s(X) \simeq G_\ell(q - d)(\kappa)[-2d]$$

is a times the truncation map of (4). The last equivalence is Gabber–Riou absolute purity. This assertion concerns maps on cohomology groups; an identity of morphisms in the derived category is not asserted.

Proof. By (3), both $T_\ell(q)(Y)$ and $T_\ell(q)(Y \setminus Z)$ lie in $D^{\geq q+1}$. Their fiber also lies in $D^{\geq q+1}$. Apply the long exact sequence to $A_\ell(q)_Z(Y) \rightarrow G_\ell(q)_Z(Y) \rightarrow T_\ell(q)_Z(Y)$.

Let $\eta_s = \mathrm{pur}_s(1) \in H^{2d}A_\ell(d)_s(X)$. By lemma 2.1, the coefficient spectrum is an E_∞ ring, including for $\ell = 2$. Purity is linear over its pullback to s , and lemma 2.1 identifies that action with the cup product with supports. Consequently, for any lift $\tilde{x} \in H^m A_\ell(w)(X)$ of $x \in H^m A_\ell(w)(\kappa)$,

$$\mathrm{pur}_s(x) = \tilde{x} \cup \eta_s, \quad b(\mathrm{pur}_s(x)) = b(\tilde{x}) \cup b(\eta_s).$$

The second equality uses the multiplicativity of b with supports. If X is henselian, every such x has a lift: for $m \leq w$ both groups are étale cohomology and henselian rigidity applies (The Stacks Project Authors, 2026, Tag 09ZI); the target is zero for $m > w$.

We first normalize $b(\eta_s)$ over the original regular local base X . Put $V = \mathbb{A}_X^d$ and consider its zero section 0_X . For the rank-one zero section in $\mathbb{A}_X^1$, smooth-pair purity and pullback invariance of H identify

$$H^2 A_\ell(1)_{0_X}(\mathbb{A}_X^1) = H^0 A_\ell(0)(X) = \mathbf{Z}/\ell.$$

The fundamental class η_0 generates this line. The comparison is injective in degree $2 = 1 + 1$ by the first assertion; therefore $b(\eta_0) = u \mathrm{Cl}_0$ for a unit u , where Cl_0 is the étale divisor class. Here and below the smooth-pair input is (Déglise et al., 2021b, arXiv version 3, Proposition 4.3.10(i), Remark 4.3.8(i), and Remark 2.5.2), and the pullback input is (Bachmann et al., 2025, Theorem 1.6(2)).

Smooth-pair purity also makes

$$L_A = H^{2d} A_\ell(d)_{0_X}(V) = \mathbf{Z}/\ell$$

a line generated by η_{0_X} . Étale absolute purity makes $L_{\mathrm{et}} = H_{0_X, \mathrm{et}}^{2d}(V, \mu_\ell^{\otimes d})$ a line generated by Cl_{0_X} . For the coordinate projections $\mathrm{pr}_h : V \rightarrow \mathbb{A}_X^1$, the product

$$c = \mathrm{pr}_1^* \eta_0 \cup \cdots \cup \mathrm{pr}_d^* \eta_0$$

has support 0_X and lies in L_A . Its image is $u^d \mathrm{pr}_1^* \mathrm{Cl}_0 \cup \cdots \cup \mathrm{pr}_d^* \mathrm{Cl}_0$, which is nonzero. Indeed cup product with each coordinate divisor class is the Gysin map for that divisor,

and is an isomorphism by absolute purity; iterating it sends $1 \in H^0(X, \mathbf{Z}/\ell)$ to this product. Thus $b : L_A \rightarrow L_{\text{et}}$ is a nonzero map between one-dimensional $\mathbf{Z}/\ell$ -modules, hence an isomorphism. Write $b(\eta_{0_X}) = v \text{Cl}_{0_X}$ with v a unit.

Pull back by the graph $\Gamma_t : X \rightarrow V$. Its square with the zero section is Tor-independent. Transverse base change for the motivic fundamental class, and base change for Riou's generalized class, give

$$\eta_s = \Gamma_t^* \eta_{0_X}, \quad \text{Cl}_s = \Gamma_t^* \text{Cl}_{0_X}, \quad b(\eta_s) = v \text{Cl}_s.$$

The conormal bundles agree in this pullback. This proves the required normalization without a product formula for motivic fundamental classes.

For nonhenselian R , use affine étale neighborhoods (X', s') of (X, s) with $\kappa(s') = \kappa$. A principal shrinking, by prime avoidance, makes $X' \times_X s = s'$. These squares are cartesian and Tor-independent, and $X' \rightarrow X$ is a morphism of the type to which purity base change applies (Déglise et al., 2021b, arXiv version 3, Theorem 3.3.2(iii) and Proposition 2.5.4(ii)). Nisnevich excision identifies their supported A_ℓ and G values with those on X . Finitariness and étale continuity identify the colimit of these values with the supported values on the henselization (Bachmann et al., 2025, Theorem 1.1(3)). A lift of x on the henselization consequently comes from a lift on one X' . Linearity, multiplicativity, and the normalization of $\eta_{s'}$ apply on that X' , and descend through excision to X . This uses no purity base-change assertion for the henselization map itself.

Finally cup product with Cl_s is the étale absolute-purity isomorphism. The comparison at κ is the truncation map by (4); take $a = v$. $\square$

Proposition 2.4 (The prime-to-characteristic defect window). *Let R be regular noetherian local of dimension $d \geq 1$, and let ℓ be a prime invertible in R . For every integer q ,*

$$\Delta_\ell(q) \in D^{[q+2, q+d]}.$$

More generally $\Delta(q)/\ell^m$ has this window for every $m \geq 1$. The lower bound uses the prime-to-characteristic comparison and étale absolute purity; the integral upper bound $\Delta(q) \in D^{\leq q+d}$ uses the value filtration. No henselian or excellence assumption and no dimension bound is used.

Proof. Gabber's purity identifies $G_\ell(q)_s(X)$ with $G_\ell(q-d)(\kappa)[-2d]$. Put $S = A_\ell(q-d)(\kappa)[-2d]$. By (4), $S = \tau^{\leq q+d} G_\ell(q)_s(X)$. By lemma 2.3, $b \circ \text{pur}_s(q)$ is an isomorphism on H^n for $n \leq q+d$ and injective in every degree. The supported comparison is an isomorphism through q and injective at $q+1$ by lemma 2.3. At $q+1 \leq q+d$ the composite is surjective, so the supported comparison is surjective as well, hence an isomorphism. Thus $\text{pur}_s(q)$ is an isomorphism through $q+1$. It is injective at $q+2$: if $d \geq 2$ this follows from the isomorphism of the composite; if $d = 1$ its source in that degree is zero. The long exact sequence gives $\Delta_\ell(q) \in D^{\geq q+2}$.

Integrally, (2) gives $A(q)(X) \in D^{\leq q+d}$ and $A(q)(U) \in D^{\leq q+d-1}$. Their fiber is in $D^{\leq q+d}$. Also $A(q-d)(\kappa)[-2d] \in D^{\leq q+d}$. Their cofiber therefore has the same upper bound. Finite coefficient cofibers preserve it. The lower bound for ℓ^m follows from the case $m = 1$ by the exact coefficient sequences $\Delta/\ell^{m-1} \rightarrow \Delta/\ell^m \rightarrow \Delta/\ell$, since $D^{\geq q+2}$ is extension-closed. $\square$

Proposition 2.5 (Absolute Adams operations and exact eigenweights). *For every integer $k \geq 1$ there is an E_∞ endomorphism*

$$\psi_{\mathbf{Z}}^k : KGL_{\mathbf{Z}}[1/k] \longrightarrow KGL_{\mathbf{Z}}[1/k]$$

such that, for the unit u and multiplication β by the Bott element,

$$\psi^k \circ u \simeq u, \quad \psi^k \circ \beta \simeq k\beta \circ \Sigma^{2,1}\psi^k. \quad (6)$$

Pullback gives these relations over every qcqs scheme, naturally under further pullback. The localization $[1/k]$ is in coefficients; k need not be invertible on the scheme.

For a regular noetherian local point and every prime $\rho \nmid k$, the operation acts on the filtered Gysin fibre Ψ_{ρ^m} for all $m \geq 1$. On its weight- q graded piece it acts by exactly k^q . Its spectral-sequence pages are $\mathbf{Z}/\rho^m$ -modules and the differentials are linear. For a differential from weight q to weight $q + r$,

$$(k^r - 1)d_r = 0. \quad (7)$$

The prime ρ need not be invertible on the local ring.

Proof. We use the absolute Adams endomorphism of $KGL[1/k]$, natural over every qcqs scheme, as formulated in (Elmanto and Morrow, 2026, arXiv v2, Appendix B.1).¹ Its unit and Bott actions can be read from the motivic Snaith presentation, which identifies KGL with the Bott localization of $\Sigma_+^\infty B\mathbf{G}_m$ over a noetherian base of finite Krull dimension (Gepner and Snaith, 2010, arXiv v3, Theorem 4.17 and Section 2.1). Delooping $x \mapsto x^k$ on $\mathbf{G}_m$ gives an E_∞ self-map of that presentation and sends its Bott element to $k\beta$. After inverting k in coefficients it extends across the Bott localization. The map is unital and multiplicative, which gives (6). The Bott convention $1 - [\mathcal{O}(-1)]$ agrees with $[\mathcal{O}(1)] - 1$ because the square of the latter is zero in $K_0(\mathbf{P}^1)$ (Bachmann et al., 2025, Definition 4.2). Finally KGL is absolute (Bachmann et al., 2025, Remark 4.3), and pullback is symmetric monoidal and exact.

On every spectrum modulo ρ^m , multiplication by k is invertible, so reduction agrees with reduction after coefficient localization $[1/k]$. Covers and slices commute with filtered colimits and cofibers. The zero slice of the unit is an equivalence (Bachmann et al., 2025, arXiv v1, Corollary 9.7). The unit relation therefore gives $s_0(\psi^k) = \text{id}$. The Bott relation and $s_q(T^q \otimes E) = T^q \otimes s_0 E$ give

$$s_q(\psi^k) = k^q \text{id} \quad \text{on } s_q(KGL[1/k]). \quad (8)$$

Naturality of purity makes

$$E \longmapsto \text{fib}(\text{map}(\mathbf{1}_s, \text{Th}(-N) \otimes i^* f_j E) / \rho^m \longrightarrow \text{map}(\mathbf{1}_s, i^! f_j E) / \rho^m)$$

an exact functor to filtered spectra. Its weight- q graded functor is an exact functor of $s_q E$. Applying it to the single homotopy (8) proves the scalar k^q on $\text{gr}^q \Psi_{\rho^m}$. In the untwisted Bott identification the source action is $k^d \psi^k$, whose weight- $q - d$ scalar is $k^d k^{q-d} = k^q$.

The integral Gysin cofiber lies in $D(\mathbf{Z})$; its derived reductions have $\mathbf{Z}/\rho^m$ -module cohomology. These are the E_1 terms, and later pages are subquotients. Their additive differentials are therefore $\mathbf{Z}/\rho^m$ -linear. They commute with the filtered operation. Comparing scalars in weights q and $q + r$ and cancelling the unit k^q proves (7). No ring structure on a Moore spectrum is used. $\square$

¹For the Snaith construction, compare (Bachmann and Hopkins, 2020, arXiv v4, Sections 3.3.1–3.3.2 and Lemmas 3.10–3.11). Their Section 3 is written under the standing hypothesis $1/2 \in S$; the general qcqs formulation used here is the one in Elmanto–Morrow. The KGL construction uses Snaith’s presentation over $\text{Spec } \mathbf{Z}$ and the Bott calculation in $K_0(\mathbf{P}^1)$; it inverts k in the coefficient spectrum and requires no inversion of 2 on the base.

Lemma 2.6 (The prime-power Adams window). *Let ℓ be prime, let N be an integer, and let $B_q \in D(\mathbf{Z})$ for $q \geq 0$. Suppose for $m = 1, 2$ there is a filtered spectrum $\mathcal{T}_m$ with $\mathcal{T}_m^0 = 0$, connectivity tending to infinity with its filtration index, and*

$$\mathrm{gr}^q \mathcal{T}_m = (B_q/\ell^m)[2q - \epsilon], \quad B_q/\ell^m \in D^{[q+2, q+N]},$$

where ϵ is one fixed integer, equal to 0 or 1. Assume the Adams relations (7) for all k prime to ℓ . Then $B_q/\ell = 0$ for every q if $N \leq \ell + 2$. If $N \leq \ell + 1$, only the hypotheses for $m = 1$ are needed. No compatibility between the two filtrations is required beyond their displayed graded identifications with reductions of the same B_q .

Proof. Write $e = n - q$ for the excess of a class in $H^n(B_q/\ell^m)$. Its homotopy degree is $i = 2q - \epsilon - n = q - \epsilon - e$. Thus only the weights $q \in [i + \epsilon + 2, i + \epsilon + N]$ contribute to a fixed degree. The finite window together with the increasing connectivity gives strong convergence of the filtered spectral sequence to $\pi_* \mathcal{T}_m^0 = 0$. For a fixed degree, truncate at a stage where the tail has zero homotopy in that degree and its neighboring degree. The finite filtration spectral sequence then computes that degree, and the finite window makes its pages stabilize as the truncation increases. A differential d_r increases weight by r and decreases homotopy degree by 1, so it increases excess by $r + 1$. A nonzero differential requires $1 \leq r \leq N - 3$.

If $N \leq \ell + 1$, choose k primitive modulo ℓ . Then $k^r - 1$ is a unit modulo ℓ for $1 \leq r \leq N - 3 \leq \ell - 2$. For $m = 1$ all differentials vanish, so $E_1 = E_\infty = 0$ and $B_q/\ell = 0$. This also covers $N \leq 3$, where there is no possible differential.

Now let $N = \ell + 2$. Choose k primitive modulo ℓ^2 if ℓ is odd, and choose $k = 3$ if $\ell = 2$. For $1 \leq r \leq \ell - 2$, $k^r - 1$ is prime to ℓ , while $k^{\ell-1} - 1 = \ell u$ with u prime to ℓ . Thus, for $m = 1, 2$, the only possible differential is $d_{\ell-1}$ and $\ell d_{\ell-1} = 0$. It runs from excess 2 in weight q to excess N in weight $q + \ell - 1$. Strong convergence to zero gives, for all q and $m = 1, 2$,

$$H^{q+e}(B_q/\ell^m) = 0 \quad (3 \leq e \leq N - 1), \quad (9)$$

$$d_{\ell-1} : H^{q+2}(B_q/\ell^m) \xrightarrow{\sim} H^{(q+\ell-1)+N}(B_{q+\ell-1}/\ell^m), \quad (10)$$

$$H^{q+N}(B_q/\ell^m) = 0 \quad (q < \ell - 1), \quad (11)$$

$$\ell H^{q+2}(B_q/\ell^m) = 0. \quad (12)$$

The isomorphism holds because the lower groups have no incoming differentials and the upper groups have no outgoing differentials. The last line uses its injectivity and $\ell d_{\ell-1} = 0$.

Fix q and put $G_n = H^n(B_q)$. The universal-coefficient sequences are

$$0 \longrightarrow G_n/\ell^m \longrightarrow H^n(B_q/\ell^m) \longrightarrow G_{n+1}[\ell^m] \longrightarrow 0. \quad (13)$$

At $n = q + 1$, $m = 1$, the window gives $G_{q+2}[\ell] = 0$. At $n = q + 2$, $m = 2$, (12) gives $\ell(G_{q+2}/\ell^2) = 0$. For $g \in G_{q+2}$ write $\ell g = \ell^2 h$; injectivity of multiplication by ℓ gives $g = \ell h$. Hence $G_{q+2} = \ell G_{q+2}$. The $m = 2$ exponent bound on the quotient term also gives $\ell G_{q+3}[\ell^2] = 0$. At $n = q + 3$, $m = 1$, (9) gives $G_{q+3}/\ell = 0$. If $\ell x = 0$, write $x = \ell y$; then $\ell^2 y = 0$, so the exponent bound gives $\ell y = 0$ and $x = 0$. Thus $G_{q+3}[\ell] = 0$. Finally (13) at $n = q + 2$, $m = 1$ gives $H^{q+2}(B_q/\ell) = 0$.

The top groups in weights at least $\ell - 1$ vanish by (10); the smaller weights vanish by (11). The intermediate groups already vanish. The whole window is therefore zero for $m = 1$. $\square$

At the endpoint this argument uses the reductions modulo ℓ and ℓ^2 ; no higher prime-power input is required.

Proof of Theorem B on values. The dimension-zero case is the identity. For $d \geq 1$, apply lemma 2.6 to

$$B_q = \Delta(q), \quad \epsilon = 1, \quad \mathcal{T}_m = \Psi_{\ell^m}, \quad N = d.$$

The bottom equivalence and increasing connectivity are lemma 2.2, the windows are proposition 2.4, and the Adams relations are proposition 2.5. Thus $\Delta(q)/\ell = 0$ for every $q \geq 0$. Negative weights vanish. The coefficient sequences $\Delta(q)/\ell^{m-1} \rightarrow \Delta(q)/\ell^m \rightarrow \Delta(q)/\ell$ then give vanishing for every m . This proves the value assertion. The passage to objects and regular pairs is proved in section 5. $\square$

Proposition 2.7 (Rational purity). *Let $i : Z \rightarrow Y$ be a regular closed immersion between regular noetherian schemes of finite dimension. Then*

$$\mathrm{pur}_i(H_Y \otimes \mathbf{Q}) : \mathrm{Th}(-N_i) \otimes i^*(H_Y \otimes \mathbf{Q}) \xrightarrow{\sim} i^!(H_Y \otimes \mathbf{Q}).$$

The integral purity cofiber for H therefore has zero rationalization. Its mapping spectrum from any compact test object has torsion homotopy groups. No finite-type hypothesis over $\mathrm{Spec} \mathbf{Z}$ is imposed.

Proof. Rationalization commutes with i^* and the Thom twist. It commutes with $i^!$ because i_* preserves compact objects, so its right adjoint preserves filtered colimits; we use the purity colimit and retract formalism of (Déglise et al., 2021b, arXiv version 3, Remark 4.3.14). The rational zero slice is the weight-zero Adams summand

$$H_{Y, \mathbf{Q}} \simeq KGL_Y^{(0)} = H\mathbf{Q}_Y$$

of $KGL_{Y, \mathbf{Q}}$ (Bachmann et al., 2025, Corollary 4.52(2), Definition 4.53, and Remark 4.54). The KGL purity transformation is invertible for these regular schemes: the Cisinski–Déglise class is invertible (Cisinski and Déglise, 2019, arXiv version 4, Theorem 13.6.3), and its match with the motivic fundamental-class purity map is the comparison in (Déglise et al., 2021b, arXiv version 3, Paragraph 4.4.3 and footnote 12); the corresponding orientation statement is (Déglise, 2018, Example 4.1.9(1) and Proposition 4.2.2). The cited comparison applies to regular noetherian finite-dimensional schemes without a finite-type hypothesis over $\mathbf{Z}$. Purity is natural in its argument, so the rational retract $KGL_Y^{(0)}$ is pure. Exactness then gives zero rationalization of the integral cofiber. Compact test objects commute with this rationalization, and a spectrum has zero rationalization exactly when all its homotopy groups are torsion. $\square$

3 Valutive polynomial invariance

Fix a prime p and $q \geq 0$. Keep $A(q)$ integral, and set

$$\begin{aligned} M_p(q) &= \mathbf{Z}/p(q)^{\mathrm{mot}}, \\ F_p(q) &= L_{\mathrm{cdh}} M_p(q) = \mathbf{Z}(q)^{\mathrm{cdh}}/p, \\ A_p(q) &= L_{\mathbb{A}^1} M_p(q) = L_{\mathbb{A}^1} F_p(q). \end{aligned}$$

Here M_p is Bouis’s theory; the identifications and localization units are (Bouis, 2024, arXiv v3, Definition 3.12, Remark 3.25, and Theorem 6.5) and (Bachmann et al., 2025, arXiv v1, Theorem 1.1(9)). Put

$$\mathcal{S}_p(q) = \mathbf{Z}/p(q)^{\mathrm{syn}}, \quad \mathcal{E}_p(q)(X) = R\Gamma_{\mathrm{\acute{e}t}}(X[1/p], \mu_p^{\otimes q}),$$

$$\begin{aligned}
Q_p(q) &= \operatorname{cofib}(F_p(q) \rightarrow A_p(q)), \\
\mathcal{N}_p(q) &= \operatorname{cofib}(M_p(q) \rightarrow F_p(q)), \\
\mathcal{D}_p(q)(X) &= \operatorname{cofib}(M_p(q)(X) \rightarrow M_p(q)(\mathbb{A}_X^1)).
\end{aligned}$$

These are cofibers, whereas all supported values are fibers. The canonical comparison $F_p \rightarrow \mathcal{E}_p$ factors through A_p compatibly with its localization unit (Bachmann et al., 2025, arXiv v1, (8.10), Theorem 7.16, and Theorem 1.13).

A valuation ring has rank equal to its Krull dimension; fields have rank zero. Mixed characteristic $(0, p)$ means that its fraction field has characteristic zero and p lies in its maximal ideal. F-smoothness is relative to p in the sense of (Bhatt and Mathew, 2023, arXiv v2, Definition 4.1). It presupposes p -quasisyntomicity: for a p -torsionfree ring, the p -completed cotangent complex has Tor-amplitude $[-1, 0]$. Its divided Frobenius fibers have p -complete Tor-amplitude at least $i + 2$, and its Nygaard filtration is complete. These conditions depend only on derived p -completion. Regular noetherian rings are F-smooth. F-smoothness passes to filtered colimits and localizations, and to a p -completely flat algebra B over T when $T/p \rightarrow B/p$ is smooth (Bhatt and Mathew, 2023, arXiv v2, Propositions 4.6, 4.9, 4.10, and Theorem 4.15). For a p -torsionfree F-smooth ring T ,

$$\operatorname{fib}(\mathcal{S}_p(q)(T) \rightarrow \mathcal{E}_p(q)(T)) \in D^{\geq q+1}. \quad (14)$$

This is (Bhatt and Mathew, 2023, arXiv v2, Proposition 5.2). The analogous bound for the fiber of $M_p \rightarrow \mathcal{E}_p$ is (Bouis, 2024, arXiv v3, Corollary 5.12); neither bound requires noetherianity.

Proposition 3.1 (Comparison, descent and Milnor excision). *For every $q \geq 0$, $Q_p(q)$ is a finitary cdh sheaf with values in $D(\mathbf{Z}/p)$ and satisfies Milnor excision. It vanishes on qcqs schemes over $\mathbf{F}_p$ or $\mathbf{Q}$, and on qcqs $\mathbf{Z}[1/p]$ -schemes of finite valuative dimension. In particular it vanishes on every ring in which p is invertible, evaluated as an affine scheme.*

On every qcqs $\mathbf{Z}[1/p]$ -scheme, $M_p(q) \rightarrow F_p(q)$ is an equivalence. On every qcqs $\mathbf{Q}$ -scheme, therefore, $M_p(q) \simeq F_p(q) \simeq A_p(q)$ and $\mathcal{D}_p(q) = 0$.

Proof. F_p and A_p are finitary cdh sheaves by (Bachmann et al., 2025, arXiv v1, Theorems 1.1(3) and 7.12(4)); a cofiber of a map of sheaves in a stable category is again a sheaf. The equal-characteristic vanishing follows from that source's Corollary 1.11. When p is invertible, the hypothesis $\operatorname{Val}(-, p, j)$ there is vacuous; Corollary 9.4 and Remark 8.2 give the stated finite-valuative-dimensional vanishing. Every noetherian scheme of finite Krull dimension has finite valuative dimension by (Elmanto et al., 2021, arXiv v3, Proposition 2.3.2(9)). An arbitrary $\mathbf{Z}[1/p]$ -algebra is the filtered union of its finitely generated subalgebras. Finitarity therefore proves the assertion for arbitrary affine schemes where p is a unit.

Milnor excision for $\mathbf{Z}(q)^{\operatorname{cdh}}$ is (Bachmann et al., 2025, arXiv v1, Theorem 7.21). The absolute motivic spectrum representing $\mathbf{Z}(q)^{\mathbb{A}^1}$ is (Bachmann et al., 2025, arXiv v1, Definition 1.5, Theorem 1.6(2), and Theorem 9.3(2)); the presheaf of its values satisfies Milnor excision on all schemes by (Elmanto et al., 2020, arXiv v2, Corollary 1.2). Reduction modulo p and taking cofibers preserve cartesian squares. Consequently, for any prime P of a valuation ring V ,

$$Q_p(q)(V) \simeq Q_p(q)(V_P) \times_{Q_p(q)(\kappa(P))} Q_p(q)(V/P). \quad (15)$$

The underlying ring square is the Milnor square of (Elmanto et al., 2021, arXiv v3, equation (3.3.1)).

The cartesian square of (Bouis, 2024, arXiv v3, Theorem B) compares $M_p \rightarrow F_p$ with the BMS syntomic theory and its cdh sheafification. That theory depends only on the derived p -completion (Bouis, 2024, arXiv v3, Remark 2.16), so both BMS terms vanish where p is invertible. The square proves the first claim. The equal-characteristic vanishing already proved, together with the polynomial invariance of A_p , proves the remaining claims. $\square$

Lemma 3.2 (Injectivity at valuation-local generic points). *Let Λ be a valuation ring, let R' be the henselization of an essentially smooth local Λ -algebra, and let s be the image of its closed point in $\mathrm{Spec} \Lambda$. The fibre of $\mathrm{Spec} R'$ over s is irreducible; write η for its generic point. Suppose G is a finitary Nisnevich sheaf of spectra on qcqs Λ -schemes and, for every qcqs T , the two maps*

$$j_T^*, \pi_T^* \infty_T^* : \pi_* G(\mathbf{P}_T^1) \longrightarrow \pi_* G(\mathbb{A}_T^1) \quad \text{agree.} \quad (16)$$

Here j_T is the standard affine chart, ∞_T its complementary section, and $\pi_T : \mathbb{A}_T^1 \rightarrow T$ the projection. Then $\pi_n G(R') \rightarrow \pi_n G(R'_\eta)$ is injective for every integer n .

In this setting, put $\Lambda' = (\Lambda_s)^h$. Then R'_η is a valuation ring with the value group of Λ' and is ind-smooth over Λ' . In particular it has rank one if Λ' has rank one, and is a field if Λ' is a field.

Let T be local and let $a \in \mathfrak{m}_T$ satisfy $V(a) = \{\mathfrak{m}_T\}$ set-theoretically. For a finitary Nisnevich sheaf G of spectra or complexes,

$$G(T) \simeq G(T[1/a]) \times_{G(T^h[1/a])} G(T^h).$$

If T is a rank-one valuation ring and $a \neq 0$, the two localized rings here are $\mathrm{Frac} T$ and $\mathrm{Frac} T^h$.

Proof. The irreducibility and the reduction replacing Λ by $(\Lambda_s)^h$ are the first claim and first reduction of (Bouis and Kundu, 2025, arXiv v2, Theorem 3.2). Thus assume that Λ is henselian and s is its closed point. We give the remaining argument at the level of fibre spectra.

Let $\sigma \in \pi_n G(R')$ vanish after localization at η . Finitarity gives $f \notin \eta$ such that $\sigma|_{R'_f} = 0$. The support sequence gives a lift of σ to $\pi_n G_{V(f)}(R')$. Expressing R' as the filtered colimit of affine Nisnevich neighbourhoods of a smooth affine finite-presentation model (X, x) , and using that filtered colimits commute with fibres, descends this lift to $\sigma_0 \in \pi_n G_Z(X)$ with $Z = V(f_0)$.

After shrinking X , its closed fibre is irreducible. The flat map from the fibre of $\mathrm{Spec} R'$ has image containing the generizations of x , so its generic point maps to the generic point of this closed fibre. As $f(\eta) \neq 0$, Z_s has positive codimension in X_s . The presentation lemma (Bouis and Kundu, 2025, arXiv v2, Corollary 2.2) gives a Nisnevich neighbourhood $X' \rightarrow X$, a smooth scheme T over a Nisnevich neighbourhood of $\mathrm{Spec} \Lambda$, and an étale map $\psi : X' \rightarrow \mathbb{A}_T^1$ such that $Z' = Z \times_X X'$ maps isomorphically to a closed subscheme $Z'' \subset \mathbb{A}_T^1$ finite over T and $Z' = X' \times_{\mathbb{A}_T^1} Z''$. Since Λ is henselian, X' can be used as a Nisnevich neighbourhood of x , with the same henselized local ring. The finite T -scheme Z'' is closed also in $\mathbf{P}_T^1$ and misses infinity.

Nisnevich excision, first for ψ and then for the open immersion $\mathbb{A}_T^1 \subset \mathbf{P}_T^1$, gives

$$G_{Z''}(\mathbf{P}_T^1) \simeq G_{Z''}(\mathbb{A}_T^1) \simeq G_{Z'}(X').$$

This uses excision on spectra, not a claim that the kernels of maps of individual homotopy presheaves satisfy excision. Let τ correspond to $\sigma_0|_{X'}$, and let $\tilde{\tau}$ be its image in $\pi_n G(\mathbf{P}_T^1)$.

Restriction to infinity is zero because the class is supported on Z'' . By (16), its restriction to $\mathbb{A}_T^1$ is zero. Pulling back to X' and then to $\text{Spec } R'$ shows $\sigma = 0$.

The generic local ring. Smoothness gives étale coordinates over $\mathbf{A}_{\Lambda'}^r$ near the point under consideration. The local ring at the generic point of the closed fibre of the affine space is

$$G_0 = \Lambda'[t_1, \dots, t_r]_{\mathfrak{m}_{\Lambda'} \Lambda'[t]}.$$

It is the Gauss valuation ring. To verify this directly, extend the valuation to polynomials by the minimum of the finitely many coefficient values. Divide a polynomial by a coefficient of smallest value; its reduction is a nonzero residue polynomial. Products of those reductions are nonzero, so the minimum rule is multiplicative. For a rational function of nonnegative value, dividing numerator and denominator by a coefficient of minimum value in the denominator makes the numerator integral and the denominator primitive. It therefore lies in the displayed localization. Conversely that localization has nonnegative value. The coefficient field already realizes all values, proving the claim about the value group.

At each affine Nisnevich neighbourhood of x , the relevant generic local ring is a localization of an étale G_0 -algebra at a prime over its maximal ideal. It is a weakly unramified extension of valuation rings by (The Stacks Project Authors, 2026, Tags 0ASJ and 0ASG). The transition maps are local and flat, hence injective. Their filtered colimit is R'_η , which is therefore a valuation ring with this same value group. The localizations and étale maps are ind-smooth over Λ' .

Henselization excision. Present T^h by pointed affine étale neighbourhoods of $\text{Spec } T$. Their closed fibres are finite discrete schemes. By localizing at an element nonzero at the distinguished point and zero at the other closed-fibre points, arrange that the fibre over the reduced closed point of $\text{Spec } T$ is exactly $\text{Spec } \kappa(T)$. Each resulting étale neighbourhood and the open $\text{Spec } T[1/a]$ form a Nisnevich square. Pass its cartesian value square to the filtered colimit. For a rank-one valuation ring the only primes are zero and the maximal ideal, and its henselization has the same value group by (The Stacks Project Authors, 2026, Tag 0ASK). $\square$

Lemma 3.3 (The rank-one local upper bound). *Let Λ be a henselian valuation ring of rank at most one with fraction field of characteristic zero, let X be smooth over Λ , and let $R' = \mathcal{O}_{X,x}^h$. Then $M_p(i)(R') \in D^{\leq i}$ for every $i \geq 0$.*

Proof. The motivic projective-line bundle formula with its natural Chern class (Bouis, 2025, arXiv v1, Theorem 4.7) shows directly that $M_p(i)$ satisfies (16): the base summand restricts by projection, and the Chern class of $\mathcal{O}(1)$ restricts to that of a trivial line bundle on both the affine chart and infinity. Thus lemma 3.2 reduces the upper bound to a field or a possibly nonhenselian rank-one valuation ring V with characteristic-zero fraction field K .

At every henselian valuation ring W , including fields, one has

$$M_p(i)(W) \simeq \tau^{\leq i} \mathcal{S}_p(i)(W). \quad (17)$$

This is (Bouis, 2026, arXiv v1, Proposition 5.3); its proof uses (Bouis, 2024, arXiv v3, Theorem 5.10) and cdh-locality of henselian valuation rings. For a field of characteristic zero, its top group is naturally

$$H^i M_p(i)(K) = H_{\text{ét}}^i(K, \mu_p^{\otimes i}) = K_i^M(K)/p,$$

by (Bouis, 2026, arXiv v1, Proposition 5.5) and (Voevodsky, 2011, Theorem 6.16). The identification uses products of unit classes and is natural in K .

Let $K^h = \text{Frac } V^h$. The square of [lemma 3.2](#) gives the upper bound except possibly in degree $i + 1$, whose group is the cokernel of

$$H^i M_p(i)(K) \oplus H^i M_p(i)(V^h) \longrightarrow H^i M_p(i)(K^h).$$

For $i = 0$ the field restriction on constant $\mathbf{Z}/p$ is an isomorphism. For $i > 0$, it suffices to show that

$$K^\times \longrightarrow (K^h)^\times / ((K^h)^\times)^p \quad \text{is surjective.} \quad (18)$$

Here is the rank-one argument for that surjectivity. The extension $V \subset V^h$ preserves the value group by ([The Stacks Project Authors, 2026](#), Tag 0ASK); hence write any $z \in (K^h)^\times$ as bu , with $b \in K^\times$ and $u \in (V^h)^\times$. The completion of the rank-one valued field K is henselian: Newton iteration at a simple residue root has errors whose values at least double at each step, and positive multiples of a fixed value are cofinal in a rank-one value group. The sequence converges to a root in the complete field. The universal property embeds K^h in this completion, with its given valuation. Thus K is dense in K^h . Choose $u_0 \in V^\times$ so that $u/u_0 = 1 + y$, with $v(y) > 2v(p)$ if $p \in \mathfrak{m}_V$, and $v(y) > 0$ if p is a unit. If p is a unit, the simple root of $Z^p - (1 + y)$ modulo the maximal ideal lifts in V^h . Otherwise the polynomial

$$g(Z) = \frac{(1 + pZ)^p - 1 - y}{p^2} \in V^h[Z]$$

reduces to Z for odd p , and to $Z + Z^2$ for $p = 2$. In both cases zero is a simple residue root. The simple-root form of Hensel's lemma applies also to nonmonic polynomials ([The Stacks Project Authors, 2026](#), Tag 04GG). For its lift w we have $(1 + pw)^p = 1 + y$, proving (18). Multilinearity of Milnor symbols makes the map of top groups surjective, so the possible group in degree $i + 1$ vanishes. $\square$

Theorem 3.4 (Polynomial invariance over rank-one valuation rings). *Let V be a henselian rank-one F -smooth valuation ring of mixed characteristic $(0, p)$. For every $q \geq 0$ and $r \geq 0$, the pullback*

$$M_p(q)(V) \longrightarrow M_p(q)(V[t_1, \dots, t_r])$$

is an equivalence. Moreover $M_p(q)(V) \simeq F_p(q)(V) \simeq A_p(q)(V)$, so $Q_p(q)(V) = 0$. The same polynomial-invariance conclusion holds for $M_p(q)(X)$ for every ind-smooth qcqs V -scheme X .

Proof. *The one-variable assertion on V .* We first prove $\mathcal{D}_p(q)(V) = 0$. Write $T^i(X) = \text{Fil}_{\text{mot}}^i K(X)/p$ for Bouis's complete motivic filtration, whose graded pieces are $M_p(i)(X)[2i]$, with $T^0(X) = K(X)/p$ ([Bouis, 2024](#), arXiv v3, Theorem C(1) and Theorem 4.46). Set

$$\Psi^i = \text{fib}(T^i(V) \longrightarrow T^i(V[t])).$$

Every valuation ring is K -regular by ([Kelly and Morrow, 2018](#), arXiv v1, Theorem 3.3(i)), so $\Psi^0 = 0$.

The upper bound of [lemma 3.3](#) at V and at all Nisnevich local rings of $V[t]$ gives $M_p(i)(V) \in D^{\leq i}$. The scheme $\text{Spec } V[t]$ has dimension at most two: its valuative dimension is two by ([Elmanto et al., 2021](#), arXiv v3, Proposition 2.3.2(7),(8)). Its Nisnevich topos has homotopy dimension at most two and is Postnikov complete by ([Clausen and Mathew, 2021](#), arXiv v3, Theorems 3.2 and 3.18). Nisnevich descent consequently gives $M_p(i)(V[t]) \in D^{\leq i+2}$.

In spectrum notation the two i th graded pieces are respectively i -connective and $(i - 2)$ -connective. This also gives

$$T^i(V) \in \text{Sp}_{\geq i}, \quad T^i(V[t]) \in \text{Sp}_{\geq i-2}, \quad \Psi^i \in \text{Sp}_{\geq i-3}. \quad (19)$$

Indeed each finite quotient T^i/T^n has the corresponding lower connectivity bound; for fixed k its π_k and π_{k+1} stabilize as n increases because the next graded piece is $(n-2)$ -connective. Completeness identifies T^i with the inverse limit of these quotients, and stabilization eliminates the possible $\lim^1$ loss. The last bound in (19) is the fibre bound, with a loss of one on the target.

Next, V and $V[t]$ are p -torsionfree, p -quasisyntomic and F -smooth by the stability results in section 3. The motivic comparison with $\mathcal{E}_p(q)$ is an isomorphism in degrees at most $q-1$ and injective in degree q on both rings. Put $K = \text{Frac } V = V[1/p]$ and $a : \text{Spec } K[t] \rightarrow \text{Spec } K$. This projection is the scheme of splittings of $0 \rightarrow \mathcal{O} \rightarrow \mathcal{O}^{\oplus 2} \rightarrow \mathcal{O} \rightarrow 0$. For $\Lambda = \mathbf{Z}/p$, the unit $\Lambda \rightarrow Ra_*\Lambda$ is an isomorphism by (Riou, 2014, Proposition 1.5, pp. 303–304). The sheaf $\mu_p^{\otimes q}$ is étale locally free of rank one over Λ , so the same statement holds after this twist. Taking derived global sections gives $\mathcal{E}_p(q)(V) \simeq \mathcal{E}_p(q)(V[t])$.

The pullback $M_p(q)(V) \rightarrow M_p(q)(V[t])$ is split by $t=0$, so its cofiber cohomology is its degreewise cokernel. It vanishes below degree q by the comparison isomorphisms. In degree q , for $w \in H^q M_p(q)(V[t])$, the difference $w - \text{pr}^*(w|_{t=0})$ maps to zero in $H^q \mathcal{E}_p(q)(V[t])$. Injectivity of the motivic comparison there makes that difference zero. Consequently

$$\mathcal{D}_p(q)(V) \in D^{\geq q+1}. \quad (20)$$

The graded fibre of Ψ is

$$\text{gr}^q \Psi = \mathcal{D}_p(q)(V)[2q-1], \quad \pi_k \text{gr}^q \Psi = H^{2q-1-k} \mathcal{D}_p(q)(V) = 0 \quad (k \geq q-1).$$

If $\Psi^q = 0$, the graded-fibre triangle identifies $\text{gr}^q \Psi \simeq \Sigma \Psi^{q+1}$. By (19) the latter is $(q-1)$ -connective. Its homotopy groups thus vanish both below $q-1$ and at or above $q-1$, so it is zero and $\Psi^{q+1} = 0$. Induction from $\Psi^0 = 0$ proves $\mathcal{D}_p(q)(V) = 0$ in every weight. This is an induction for all integers $q \geq 0$, not a finite test of the indices.

Smooth schemes and polynomial extensions. The defect $\mathcal{D}_p(q)$ is a finitary Nisnevich sheaf. The natural projective-line bundle formula for M_p induces one for $\mathcal{D}_p$ by taking cofibers of the formulas on X and $\mathbb{A}_X^1$. The Chern class on the second summand restricts to zero on the affine chart and on infinity; thus it satisfies (16).

Take a smooth finite-presentation V -scheme X and a Nisnevich local ring $R' = \mathcal{O}_{X,x}^h$. If x lies over the generic point, $\mathcal{D}_p(q)(R') = 0$ by proposition 3.1. Otherwise lemma 3.2 injects every homotopy group of $\mathcal{D}_p(q)(R')$ into the corresponding group of $\mathcal{D}_p(q)(V_\eta)$, where $V_\eta = R'_\eta$. By lemma 3.2, this is a rank-one valuation ring, ind-smooth over V . Its henselization remains rank one and ind-smooth, with characteristic-zero fraction field. It is p -torsionfree, p -quasisyntomic and F -smooth. Thus the rank-one argument above kills $\mathcal{D}_p(q)(V_\eta^h)$. The two fraction-field terms in lemma 3.2 vanish by proposition 3.1. That square kills $\mathcal{D}_p(q)(V_\eta)$, proving that all Nisnevich stalks on X vanish.

A smooth finite-presentation scheme over a rank-one valuation ring has finite Krull dimension, by the valuative dimension bound for finite-type morphisms in (Elmanto et al., 2021, arXiv v3, Proposition 2.3.5 and Corollary 2.3.3). The Nisnevich topos is therefore Postnikov complete by the Clausen–Mathew results cited above. Vanishing of all its stalks gives $\mathcal{D}_p(q)(X) = 0$. Finitarity extends this to ind-smooth schemes. Applying it successively to $\mathbf{A}_V^{r-1}$ proves the displayed assertion.

The presheaf-level $\mathbf{A}^1$ -localization is the Suslin realization

$$A_p(q)(V) = \text{colim}_{[n] \in \Delta^{\text{op}}} M_p(q)(V \times \Delta^n), \quad \Delta_V^n \simeq \mathbf{A}_V^n.$$

This formula is used in (Bachmann et al., 2025, arXiv v1, proof of Corollary 9.5). All its augmentations are now equivalences, compatibly with the simplicial structure, so

$M_p(q)(V) \rightarrow A_p(q)(V)$ is an equivalence. Henselian valuation rings are cdh-local, and a finitary presheaf retains its value there under cdh sheafification; see (Bouis, 2026, arXiv v1, proof of Proposition 5.3). Hence $M_p(q)(V) \rightarrow F_p(q)(V)$ is an equivalence. Compatibility of the localization units identifies their composite with $M_p(q)(V) \rightarrow A_p(q)(V)$, proving the result. $\square$

Proposition 3.5 (Reduction to the rank-one part at p). *Let V be a mixed-characteristic valuation ring, of arbitrary rank. Let $P = \sqrt{pV}$, let J be the union of its primes not containing p , and put*

$$T = V_P, \quad W_V = T/JT.$$

Then P is prime, J is the largest prime not containing p , and there is no prime strictly between J and P . Moreover

$$JT = \bigcap_{n \geq 1} p^n T,$$

and W_V is a rank-one mixed-characteristic valuation ring. If V is F -smooth, so is W_V . If V is henselian, then T and W_V are henselian.

For a henselian valuation ring V of any rank, $Q_p(q)(V) = 0$ if $p = 0$ or p is a unit in V . Otherwise

$$Q_p(q)(V) \simeq Q_p(q)(W_V).$$

Consequently $Q_p(q)(V) = 0$ in every weight whenever such a V is F -smooth, by theorem 3.4. There is no induction on the rank and no finite-rank restriction.

Proof. Ideals in a valuation ring are totally ordered. The radical of any proper ideal is prime: if a product belongs to that radical, comparison of the principal ideals of its two factors puts a power of one factor in the original ideal. This applies to pV . The union defining J is a prime ideal and omits p . Every prime omitting p is contained in J , and every prime containing p contains P , proving adjacency.

Use the valuation on T and put $I = \bigcap_n p^n T$. If $x, y \notin I$, some a, b satisfy $v(x) < av(p)$ and $v(y) < bv(p)$; then $v(xy) < (a+b)v(p)$, so I is prime. It does not contain p . If $\mathfrak{q}$ is a prime of T omitting p and $x \in \mathfrak{q}$ were outside I , then $p^n \in xT$ for some n , contrary to $p \notin \mathfrak{q}$. Thus I is the largest prime of T omitting p , namely JT . Quotienting T by this prime leaves exactly two primes. Its fraction field is $\kappa(J)$, of characteristic zero because $p \notin J$ and every integer prime to p was already a unit. This proves the rank and characteristic claims.

If V is F -smooth, its localization T is F -smooth. As $JT \subset p^n T$ for every n , $T/p^n \simeq W_V/p^n$. Both rings are p -torsionfree. The derived p -completions therefore agree. Their cotangent complexes modulo p agree over $\mathbf{F}_p$, so p -quasisyntomicity passes from T to W_V ; the defining prismatic conditions then pass because they depend only on this completion.

For the henselian assertion we prove the required localization statement directly, using the simple-root criterion (The Stacks Project Authors, 2026, Tag 04GG(2)). For any prime $\mathfrak{p}$ of a valuation ring V , one has $\mathfrak{p}V_{\mathfrak{p}} = \mathfrak{p}$ inside $\text{Frac } V$: if $x \in \mathfrak{p}$ and $s \notin \mathfrak{p}$, comparison of xV and sV gives $x/s \in \mathfrak{p}$. Given $f \in V_{\mathfrak{p}}[X]$ with a simple residue root, translate a lift of that root to zero and divide by $f'(0)$. The resulting polynomial is $h(X) = a_0 + X + \sum_{i \geq 2} a_i X^i$, with $a_0 \in \mathfrak{p}$. Choose $c \in V \setminus \mathfrak{p}$ clearing the finitely many denominators of the a_i . Then

$$g(X) = h(cX)/c = a_0/c + X + \sum_{i \geq 2} a_i c^{i-1} X^i \in V[X].$$

Here $g(0) \in \mathfrak{p}$ and $g'(0) = 1$. Henselianity of V gives $b \in \mathfrak{m}_V$ with $g(b) = 0$. The coefficient of b in this equation is $1 + \sum_{i \geq 2} a_i c^{i-1} b^{i-1} \in V^\times$, so $b \in \mathfrak{p}$. Thus cb gives the required root

modulo $\mathfrak{p}$ in $V_{\mathfrak{p}}$, proving that every such localization is henselian. Apply this with $\mathfrak{p} = P$ to obtain henselianity of T . The same simple-root criterion passes to every quotient of a henselian local ring, and hence gives henselianity of $W_V = T/JT$. It also gives henselianity of every induced residue valuation $V/\mathfrak{p}$.

The first cases are [proposition 3.1](#). In the remaining case the Milnor square at P has two lower terms V/P and $\kappa(P)$ of characteristic p , where Q_p vanishes. It gives $Q_p(V) \simeq Q_p(V_P)$. The Milnor square for V_P at JV_P has localization V_J and residue field $\kappa(J)$, both $\mathbf{Q}$ -algebras, where Q_p also vanishes. This gives $Q_p(V_P) \simeq Q_p(W_V)$. Apply [theorem 3.4](#) when V is F -smooth. $\square$

4 Residue-characteristic comparison

We first prove the regular-surface comparison without excellence. The excellent dimension-three argument then uses regular models and two separate descent constructions: finite supports for Q_p , and bounded-below cdh detection for the étale comparison cofiber.

4.1 Regular surfaces

Lemma 4.1 (The noetherian support filtration). *Let G be a finitary Nisnevich sheaf of spectra on qcqs schemes, and let Y be a noetherian scheme of dimension at most d , with $d \geq 1$. Write*

$$h(y) = \dim \mathcal{O}_{Y,y}, \quad Y^{(r)} = \{y \in Y : h(y) = r\}, \quad X_y = \operatorname{Spec} \mathcal{O}_{Y,y}^h, \quad U_y = X_y \setminus \{y\}.$$

For $0 \leq r \leq d+1$, put

$$G^{(r)}(Y) = \operatorname{colim}_{Z \in \mathcal{Z}_r(Y)} G_Z(Y), \quad \mathcal{Z}_r(Y) = \{Z \subset Y \text{ closed} : h(z) \geq r \text{ for every } z \in Z\},$$

where supports are ordered by inclusion. There are natural equivalences

$$\operatorname{cofib}(G^{(r+1)}(Y) \longrightarrow G^{(r)}(Y)) \simeq \bigoplus_{y \in Y^{(r)}} G_{\{y\}}(X_y) \quad (0 \leq r \leq d). \quad (21)$$

Assume that

$$G_{\{y\}}(X_y) = 0 \quad (h(y) < d), \quad G(U_y) = 0 \quad (h(y) = d). \quad (22)$$

Then the map induced by finite closed supports is an equivalence

$$\bigoplus_{y \in Y^{(d)}} G(X_y) \xrightarrow{\sim} G(Y). \quad (23)$$

Every point of $Y^{(d)}$ is closed, and each homotopy class has finitely many nonzero components. Its component at y is its restriction to X_y . The lower-height condition in [\(22\)](#) follows, in particular, from $G(X_y) = G(U_y) = 0$ at those points. No excellence or catenarity assumption is made.

Proof. The support posets are filtered, since they are closed under finite unions. We have $G^{(0)}(Y) = G(Y)$ and $G^{(d+1)}(Y) = 0$. If z is a proper specialization of y , a prime chain ending at y can be extended by z ; hence $h(z) > h(y)$. This elementary inequality is the only dimension property needed for the filtration.

For closed supports $W \subset Z$, the definition of supported values gives a fibre sequence

$$G_W(Y) \longrightarrow G_Z(Y) \longrightarrow G_{Z \setminus W}(Y \setminus W).$$

Filtered colimits preserve this sequence. To calculate the layer at height r , one may therefore use pairs $W \in \mathcal{Z}_{r+1}(Y)$, $Z \in \mathcal{Z}_r(Y)$, with $W \subset Z$: replacing any pair by $(W, W \cup Z)$ gives this cofinal indexing.

A fixed Z has finitely many irreducible components. Include in W its components whose generic points have height greater than r , and the intersections of the remaining components. These subsets have height at least $r+1$. The remaining components of $Z \setminus W$ are disjoint closed supports in $Y \setminus W$, with generic points in $Y^{(r)}$. Excision splits their supported spectra as a finite direct sum. Enlarging W by proper closed subsets of these components removes all their non-generic points in the filtered limit. Zariski excision allows the ambient neighbourhood of each generic point to shrink independently. Finitarity then identifies the resulting value with

$$G_{\{y\}}(\mathrm{Spec} \mathcal{O}_{Y,y}).$$

Every $y \in Y^{(r)}$ occurs by using its closure as a support, and the filtered union of these finite sums is the direct sum over $Y^{(r)}$.

For completeness, the supported henselization identification uses pointed affine étale neighbourhoods of the local scheme $\mathrm{Spec} \mathcal{O}_{Y,y}$. Their closed fibres are finite discrete schemes. A principal shrinking keeps the distinguished point, with residue field $\kappa(y)$, and removes the other closed-fibre points. Nisnevich excision identifies its supported value with that on the original local scheme. Passing to the filtered limit gives $G_{\{y\}}(X_y)$. Complements are quasi-compact, and their inverse systems have affine transition maps, so finitarity applies also to these supported values. This proves (21) on spectra.

The vanishing assumptions kill every layer below d . At height d , $G_{\{y\}}(X_y) = G(X_y)$ because $G(U_y) = 0$, so the finite tower gives (23). A point of height d has no proper specialization in Y and is closed. A closed support consisting of such points is a finite set, by noetherianity. Excision gives the map in the displayed equivalence and identifies its components with the corresponding restrictions. Finally, homotopy groups commute with direct sums of spectra, which proves the assertion about homotopy classes. $\square$

In particular, if G vanishes on all excellent noetherian schemes of dimension less than d , the lemma applies to every excellent Y of dimension at most d . Its henselized local schemes are excellent, and their punctures have strictly smaller dimension. The same assertion holds without excellence when vanishing is known on the larger class of all lower-dimensional noetherian schemes.

For the surface comparison, M_p is a finitary Nisnevich sheaf (Bouis, 2024, arXiv v3, Theorem C), and $\mathcal{N}_p$ has the same properties. On a henselian valuation ring, the canonical map $M_p \rightarrow F_p$ is an equivalence, by cdh locality and finitarity (Bouis, 2026, arXiv v1, proof of Proposition 5.3). The regular blow-up formula for M_p holds for every regular closed immersion of qcqs schemes (Bouis, 2025, arXiv v1, Theorem 4.18).

Theorem 4.2 (Comparison on regular surfaces). *Let R be a henselian regular noetherian local ring of dimension at most two, with $0 \neq p \in \mathfrak{m}_R$. Then, for every $q \geq 0$, the canonical maps*

$$M_p(q)(R) \xrightarrow{\sim} F_p(q)(R) \xrightarrow{\sim} A_p(q)(R)$$

are equivalences. Excellence, restrictions on the residue field, and restrictions on ramification are unnecessary. The first equivalence uses the valuation and regular blow-up inputs for M_p , and does not use theorem 3.4 or rank reduction.

Proof. Fix p, q , and write G for either $\mathcal{N}_p(q)$ or $Q_p(q)$. Both are finitary Nisnevich sheaves. The preceding valuation comparison kills $\mathcal{N}_p$ on fields and henselian DVRs. For Q_p , fields

are killed by [proposition 3.1](#). A henselian DVR in which $p = 0$ or p is a unit is killed by the same proposition. In the remaining case it is mixed characteristic and F-smooth, so [theorem 3.4](#) kills its Q_p -value. Thus G vanishes on fields and henselian DVRs.

Apply [lemma 4.1](#) with $d = 1$ to a regular noetherian scheme of dimension at most one. Its henselized local rings are fields or DVRs, and the puncture of a DVR is its fraction field. The hypotheses hold and every top local value is zero. Consequently

$$G(T) = 0 \quad \text{for every regular noetherian } T \text{ with } \dim T \leq 1. \quad (24)$$

On a regular noetherian scheme Y of dimension at most two, the lower-height local support spectra vanish for the same reason. The puncture of every height-two henselized local ring is regular noetherian of dimension at most one, so (24) kills it. Hence

$$G(Y) \simeq \bigoplus_{y \in Y^{(2)}} G(\mathcal{O}_{Y,y}^h). \quad (25)$$

In particular, for every two-dimensional regular noetherian local ring T , the natural map $G(T) \rightarrow G(T^h)$ is an equivalence: its puncture has zero value, and supported Nisnevich excision applies.

Let T be such a local ring with $0 \neq p \in \mathfrak{m}_T$, and let $B_T = \mathrm{Bl}_{\mathfrak{m}_T} \mathrm{Spec} T$ and $E_T = \mathbf{P}_{\kappa(T)}^1$. The blow-up is regular of dimension two. Every closed point lies over the closed point of T , by properness, and is a closed point of E_T , with a finite residue extension and local dimension two. The cdh blow-up square gives $Q_p(T) \simeq Q_p(B_T)$, since $Q_p(E_T) = Q_p(\kappa(T)) = 0$. The centre is a regular closed immersion, so the regular blow-up formula for M_p , together with cdh excision for F_p , gives the same cartesian square for $\mathcal{N}_p$. Its centre and exceptional curve have zero $\mathcal{N}_p$ -values by (24). Thus for either G ,

$$G(T) \simeq G(B_T) \simeq \bigoplus_{b \in E_T \text{ closed}} G(\mathcal{O}_{B_T,b}^h). \quad (26)$$

The components are restrictions, and each class has finitely many nonzero components.

The case $\dim R = 1$ has already been proved. Suppose $\dim R = 2$ and choose a nonzero class $\theta \in H^j G(R)$. Formula (26) supplies a closed point with a nonzero component. Applying the same formula repeatedly gives an infinite sequence of ordinary local quadratic transforms

$$R = T_0 \subsetneq T_1 \subsetneq T_2 \subsetneq \cdots \subset \mathrm{Frac} R$$

at closed exceptional points. The selected images in $H^j G(T_i^h)$ are compatible and nonzero at every stage. Equivalently, these are the nonzero vertices along a branch of the finitely branching tree of components. The equivalence $G(T_i) \rightarrow G(T_i^h)$ ensures that the branch can be taken among the ordinary quadratic transforms; no finite-type model of R is required.

Put $V = \bigcup_i T_i$. The union is a valuation ring of $\mathrm{Frac} R$ dominating R , by ([Lévy-Bruhl, 1957–1958](#), Lemma 5, printed pp. 19-9–19-10). That lemma is stated for arbitrary two-dimensional regular local rings and successive quadratic transforms with a common fraction field. Its proof uses the general Lemma 3 and Property II in the same exposition. The algebraic or absolute hypotheses imposed in the later Property III are not hypotheses of Lemma 5. This citation refers to the published exposition and proof, rather than to an unconsulted locator in Abhyankar’s original article.

Henselization commutes with these filtered colimits along local maps:

$$V^h \simeq \mathrm{colim}_i T_i^h.$$

Indeed, a filtered colimit of henselian local rings along local maps is henselian by the finite-data simple-root criterion, and the universal property of henselization gives the displayed identity. The henselization of a valuation ring is a valuation ring ([The Stacks Project Authors, 2026](#), Tag 0ASK). It is mixed characteristic, because V dominates R and has fraction field $\text{Frac } R$. Each T_i^h is regular noetherian, hence F -smooth and p -torsionfree; the filtered colimit is F -smooth and p -torsionfree. Finitarity gives

$$G(V^h) \simeq \text{colim}_i G(T_i^h).$$

The image of θ is nonzero in this value: a class vanishes in a filtered colimit of homotopy groups only if it vanishes at some later stage, whereas the selected images are nonzero at every stage.

For $G = \mathcal{N}_p$, the henselian valuation comparison gives $G(V^h) = 0$. For $G = Q_p$, [proposition 3.5](#), using [theorem 3.4](#), gives $G(V^h) = 0$, since V^h is F -smooth. This contradicts the surviving class in either case. Therefore $\mathcal{N}_p(q)(R) = Q_p(q)(R) = 0$, proving both comparisons. The all-rank reduction makes a separate valuation-rank bound or equality case of Abhyankar's inequality unnecessary here. $\square$

4.2 Regular models and excellent-scheme descent

We use resolution of integral excellent schemes in dimensions at most two: finite normalization for curves and ([Lipman, 1978](#), Theorem, p. 151) for surfaces. In dimension three we invoke Cossart–Piltant's resolution theorem: a reduced separated noetherian quasi-excellent scheme of dimension at most three has a proper birational regular model, isomorphic over its regular locus ([Cossart and Piltant, 2019](#), arXiv v2, Theorem 1.1, p. 3). There is no residue-field or characteristic restriction, and no assertion that the model is obtained by blow-ups in regular centres is needed.

For a separated noetherian X , let Sch_X be its separated finite-type schemes. When X is excellent of finite dimension, for $d \geq 0$ define $\mathcal{W}_d(X)$ to consist of henselian valuation rings W mapping to an integral $Y' \in \text{Sch}_X$ of dimension at most d , with generic point mapping to the generic point of Y' and $\text{Frac } W$ algebraic over $\kappa(Y')$. A *trdeg-zero point* of Y is a map $\text{Spec } W \rightarrow Y$ with W henselian valuation and $\text{Frac } W$ algebraic over the residue field of the image of its generic point. It belongs to $\mathcal{W}_{\dim Y}(X)$ by taking the reduced closure of that image. There is no bound on rank or residue extension.

Proposition 4.3 (*F-smoothness from regular models*). *Let $d \leq 3$, invoking Cossart–Piltant's resolution theorem when $d = 3$. For every mixed-characteristic $W \in \mathcal{W}_d(X)$, W is a filtered union of regular local rings of dimension at most d , dominated by W and essentially of finite type over X . Thus W is p -torsionfree, p -quasisyntomic and F -smooth, and the syntomic cofiber at W belongs to $D^{\geq q}$ for all $q \geq 0$.*

Proof. Choose Y' in the definition, put $K = \kappa(Y')$ and $L = \text{Frac } W$, and choose an affine neighbourhood $U = \text{Spec } R_U \subset Y'$ of the centre of W , over an affine $\text{Spec } B \subset X$. It contains the image of $\text{Spec } W$, and $R_U \rightarrow W$ is injective. For a finite subset $T \subset W$, set

$$A_T = (R_U[T])_{\mathfrak{m}_W \cap R_U[T]} \subset W.$$

Its unlocalized ring is an integral excellent finite-type B -algebra with fraction field $K(T)$ finite over K . The dimension formula for excellent schemes gives $\dim R_U[T] \leq \dim R_U \leq d$ ([The Stacks Project Authors, 2026](#), Tag 02IJ). The A_T are a directed family with union W .

Resolve $\text{Spec } R_U[T]$. In dimension one its finite normalization is regular; finiteness follows from excellence. In dimension two Lipman's theorem applies to the excellent surface.

In dimension three use Cossart–Piltant’s resolution theorem. Take the integral component of the regular model birational to this integral scheme. The valuative criterion lifts $\text{Spec } W$ to the proper model; at its centre c_T , the regular local ring $\mathcal{O}_{X_T, c_T}$ is a subring of W dominated by W , and contains A_T . Its dimension is at most d by the same dimension formula.

The regular local subrings so obtained can be made directed. Given two of them, choose finite sets of their generators as localizations of finite-type B -algebras and include these, as well as generators of R_U , in T . The ring $\mathcal{O}_{X_T, c_T}$ then contains both: every element inverted in either localization is outside $\mathfrak{m}_W$ and hence invertible in this local ring. Their union contains all A_T and equals W .

Each regular local ring is p -quasisyntomic and F -smooth by (Bhatt and Mathew, 2023, arXiv v2, Theorem 4.15 and its proof). These properties pass to the filtered union; p -torsionfreeness also passes, or follows from the characteristic-zero fraction field of W . Apply (14) and shift from fibre to cofiber. $\square$

Theorem 4.4 (Comparison on excellent schemes of dimension at most three). *Let Y be an excellent noetherian scheme of dimension at most three. The dimension-three proof invokes Cossart–Piltant’s resolution theorem. Then $Q_p(q)(Y) = 0$ for every prime p and every $q \geq 0$. Thus $F_p(q)(Y) \rightarrow A_p(q)(Y)$ is an equivalence. The assertion allows singularities, arbitrary residue fields, and arbitrary ramification.*

Proof. Fix p, q . Let $\text{Claim}(d)$ assert the vanishing for all excellent noetherian schemes of dimension at most d . For $d = 0$, nilinvariance of a cdh sheaf reduces to a finite disjoint union of fields. The equal-characteristic and p -unit cases of proposition 3.1 kill them.

Assume $\text{Claim}(d-1)$, with $1 \leq d \leq 3$. We may first treat separated schemes: finite affine coverings and their intersections then extend the answer to arbitrary excellent noetherian schemes by Nisnevich descent. Each such intersection is an open subscheme of an affine scheme, hence is separated and has the same dimension bound. Apply lemma 4.1 to Q_p and Y . It suffices to kill $Q_p(R)$ for excellent henselian local R of dimension d .

Nilinvariance and abstract blow-up excision for a closed cover reduce this ring to a domain. Explicitly, if the reduced scheme is a union of its finitely many irreducible components, their pairwise intersections have dimension less than d whenever a component has dimension d ; components of smaller dimension are already zero. Induction on the number of components splits $Q_p(R)$ as the sum of the d -dimensional domain values. If $p = 0$ or p is a unit, proposition 3.1 finishes. Thus take a domain R with $0 \neq p \in \mathfrak{m}_R$, fraction field K of characteristic zero, and closed point s .

For every nonzero ideal $I \subset R$, its blow-up Y_I is integral, excellent, proper and birational over R . Its dimension is at most d by the dimension formula. The centre $Z_I = V(I)$ and its inverse image E_I have dimension less than d : the latter is an effective Cartier divisor when the centre is nonempty. Abstract blow-up excision and $\text{Claim}(d-1)$ give

$$Q_p(R) \simeq Q_p(Y_I). \quad (27)$$

The closed points of Y_I lie over s by properness and have residue fields finite over $\kappa(R)$. Universal catenarity and the birational dimension formula give height d at these points and height less than d elsewhere. Hence lemma 4.1 gives

$$Q_p(R) \simeq \bigoplus_{y \in Y_I \text{ closed}} Q_p(\mathcal{O}_{Y_I, y}^h).$$

Suppose a nonzero class $\theta \in H^j Q_p(R)$ exists. Let $S(Y_I)$ consist of the closed points where its component is nonzero. These are nonempty finite sets. The blow-ups of all

nonzero ideals form a directed system: blowing up IJ dominates the blow-ups of I and of J . Indeed an invertible product of nonzero ideals on an integral scheme makes each factor invertible, and the universal property of blow-ups applies. Birational dominating morphisms are unique.

Restriction induces maps $S(Y_J) \rightarrow S(Y_I)$ along every dominating morphism $Y_J \rightarrow Y_I$. They are well defined: the pullback of a class supported at a finite set has support in its inverse image, and its restriction at a point over y is the restriction of the component at y along the local map of henselizations. In particular a zero component cannot acquire a nonzero pullback component. This argument does not require surjectivity of these maps. The inverse limit of these nonempty finite sets is nonempty. One proof is to put the discrete topology on each finite set; in the compact product, the finitely many compatibility conditions in any finite collection can be satisfied by choosing a point at a common dominating blow-up. The closed conditions have the finite intersection property.

Choose a compatible family (y_I) and put

$$V = \operatorname{colim}_I \mathcal{O}_{Y_I, y_I} \subset K.$$

This is a local subring dominating R . It is a valuation ring: for $f = a/b \in K^\times$, with $a, b \in R \setminus 0$, the ideal (a, b) is principal at the chosen centre on its blow-up, so f or f^{-1} is in its local ring. The residue field of V is algebraic over $\kappa(R)$. Henselization commutes with filtered colimits of local rings along local maps: a map to a henselian local target factors uniquely through each henselization, and the filtered colimit of henselian local rings is henselian by the finite-data simple-root criterion. Thus

$$V^h = \operatorname{colim}_I \mathcal{O}_{Y_I, y_I}^h.$$

It is a valuation ring by (The Stacks Project Authors, 2026, Tag 0ASK), and its fraction field is algebraic over K , since henselization is ind-étale. Consequently $V^h \in \mathcal{W}_d(\operatorname{Spec} R)$ and is mixed characteristic. By finitariness, the image of θ in $H^j Q_p(V^h)$ is nonzero: along the selected branch every subsequent component is nonzero, so the class is never killed in the filtered colimit.

But [proposition 4.3](#) makes V^h F -smooth, using excellent-surface resolution for $d \leq 2$ and Cossart–Piltant’s resolution theorem for $d = 3$. [Proposition 3.5](#) then kills $Q_p(V^h)$, a contradiction. This proves [Claim\(d\)](#) and completes the induction. At no step is polynomial extension of an excellent d -dimensional ring used to produce a regular model. $\square$

4.3 The independent cdh comparison bound

The following argument concerns

$$C_p(q) = \operatorname{cofib}(F_p(q) \rightarrow \mathcal{E}_p(q)), \quad \operatorname{fib}(F_p(q) \rightarrow \mathcal{E}_p(q)) = C_p(q)[-1]. \quad (28)$$

It does not use valuation polynomial invariance or the preceding vanishing theorem for Q_p .

Lemma 4.5 (Trdeg-zero detection of cdh covers). *Let X be separated noetherian, and let $Y \in \operatorname{Sch}_X$. A sieve on Y is a cdh covering sieve if every trdeg-0 point of Y factors through a member of the sieve.*

Proof. By (Gabber and Kelly, 2015, arXiv v2, Theorems 2.3 and 2.6, and Table 1), spectra of henselian valuation rings give a conservative family of cdh points on Sch_X . Coverings have finite refinements because the schemes are noetherian. It suffices to lift every map $g : \operatorname{Spec} W \rightarrow Y$ from a henselian valuation ring, of arbitrary rank and fraction field.

Put $y = g(\eta_W)$, and let K_0 be the algebraic closure of $\kappa(y)$ inside $L = \text{Frac } W$. The intersection $W_0 = W \cap K_0$ is a valuation ring. Choose an affine neighbourhood of the image of the closed point of W in Y ; it contains the whole image. Its coordinate map has image in $\kappa(y) \cap W \subset W_0$, so g factors through $\text{Spec } W_0$. The inclusion $W_0 \rightarrow W$ is local and W is henselian, so it extends to $W_0^h \rightarrow W$. The henselization is a valuation ring by (The Stacks Project Authors, 2026, Tag 0ASK), with fraction field algebraic over K_0 and thus over $\kappa(y)$. The resulting map $\text{Spec } W_0^h \rightarrow Y$ is a trdeg-0 point. A lift of it through the sieve supplies a lift of g . $\square$

Proposition 4.6 (Cdh detection with a uniform lower bound). *Let X be separated excellent noetherian of finite dimension, let $d \geq 0$, and let G be a finitary cdh sheaf of complexes on qcqs X -schemes. Suppose $G(U) \in D^{\geq m_0}$ for every $U \in \text{Sch}_X$, for one integer m_0 . Then the following hold.*

1. *If $G(W) \in D^{\geq n}$ for every $W \in \mathcal{W}_d(X)$, then $G(Y) \in D^{\geq n}$ for every $Y \in \text{Sch}_X$ with $\dim Y \leq d$.*
2. *If $H^a G(U) = 0$ for all $a < j$ and all such U of dimension at most d , then for every $Y \in \text{Sch}_X$ with $\dim Y \leq d$, a class in $H^j G(Y)$ is zero if and only if its pullback is zero at every trdeg-0 point of Y .*

Proof. Let $(H_{<j})$ denote the vanishing in the second assertion. It holds initially for $j \leq m_0$. Under $(H_{<j})$, prove the second assertion by induction on $e = \dim Y$ and on the number of its irreducible components. Nilinvariance permits reduction to reduced Y . For a closed cover by one component Y_1 and the union Y_2 of the others, cdh excision gives

$$H^{j-1}G(Y_1 \cap Y_2) \longrightarrow H^j G(Y) \longrightarrow H^j G(Y_1) \oplus H^j G(Y_2).$$

The first group is zero by $(H_{<j})$. Restricting the trdeg-0 point condition to the components therefore reduces to the integral case. At dimension zero the identity of its field is itself a trdeg-0 point, so the assertion is immediate.

Let Y be integral of dimension $e > 0$, and let $c \in H^j G(Y)$ have zero pullbacks at its trdeg-0 points. Finitarity says that each of these points factors through some finite-type affine scheme $U \rightarrow Y$ where c becomes zero: express its valuation ring as the union of finite-type subalgebras over an affine neighbourhood in Y , then use vanishing of the class in the filtered colimit. The sieve of all such morphisms is covering by lemma 4.5. A finite refinement has the form

$$\{V_k \longrightarrow Y' \longrightarrow Y\},$$

with $Y' \rightarrow Y$ a proper cdh cover and $\{V_k \rightarrow Y'\}$ a Nisnevich cover. This refinement statement is (Suslin and Voevodsky, 2000, Proposition 5.9 and Section 12, first paragraph), where Section 12 extends it from fields to arbitrary noetherian bases.

The generic point has a lift with its residue field unchanged, since a proper cdh cover is completely decomposed. The reduced closure Y'_0 of that lift is proper and birational over Y , and has dimension at most e by excellence and the dimension formula. It is an isomorphism outside a nowhere dense closed $Z \subset Y$; write $E = Y'_0 \times_Y Z$. Both Z and E have dimension less than e . The abstract blow-up sequence gives

$$H^{j-1}G(E) \longrightarrow H^j G(Y) \longrightarrow H^j G(Y'_0) \oplus H^j G(Z).$$

Its first term is zero, and the component of c on Z is zero by induction on dimension. On Y'_0 , the pulled-back Nisnevich cover has Čech terms étale over Y'_0 , all of dimension at most e . The descent spectral sequence

$$E_1^{a,b} = H^b G(V'^{(a)}) \implies H^{a+b} G(Y'_0)$$

converges with finitely many terms contributing to each total degree, because $b \geq m_0$. The restriction of c to its $a = 0$ term is zero. For $a \geq 1$, the terms contributing to degree j have $b = j - a < j$ and vanish by $(H_{<j})$. Thus its restriction to Y'_0 is zero, and excision kills c . This proves the second assertion.

For the first assertion, every trdeg-0 point of a scheme of dimension at most d belongs to $\mathcal{W}_d(X)$. For $j < n$ its $H^j G(W)$ is zero. The second assertion therefore advances $(H_{<j})$ to $(H_{<j+1})$. Starting at m_0 and inducting to n proves the bound. If $n \leq m_0$ it already holds. $\square$

Lemma 4.7 (The comparison cofiber on bounded-dimensional points). *The $C_p(q)$ of (28) is a finitary cdh sheaf and satisfies $C_p(q)(U) \in D^{\geq -1}$ for every qcqs U . Let X be separated excellent noetherian of finite dimension, and let $d \leq 3$, invoking Cossart–Piltant’s resolution theorem when $d = 3$. For every $Y \in \text{Sch}_X$ of dimension at most d ,*

$$C_p(q)(Y) \in D^{\geq q-1},$$

and the restriction map

$$H^{q-1}C_p(q)(Y) \longrightarrow \prod_{y \in Y_{\mathbf{F}_p}} H^{q-1}C_p(q)(\kappa(y)) \quad \text{is injective.} \quad (29)$$

Here $Y_{\mathbf{F}_p}$ means all points where the residue characteristic is p , with no perfection assumption.

Proof. The sheaf and finitary assertions for $C_p(q)$ follow from those of $F_p(q)$ and étale cohomology; they are used in (Bachmann et al., 2025, arXiv v1, proof of Lemma 8.11(1)). The uniform bound follows because $F_p(q)$ has no negative cohomology by that source’s Lemma 7.31 and $\mathcal{E}_p(q)$ has no negative cohomology.

At a henselian valuation ring W , cdh sheafification does not change the finitary syntomic values, and the comparison factorization of (28) gives a triangle

$$T_q(W) \longrightarrow C_p(q)(W) \longrightarrow \text{cofib}(\mathcal{S}_p(q)(W) \longrightarrow \mathcal{E}_p(q)(W)), \quad T_q(W) \in D^{\geq q+1}. \quad (30)$$

Here $T_q(W) = \text{cofib}(\tau^{\leq q}\mathcal{S}_p(q)(W) \rightarrow \mathcal{S}_p(q)(W))$. This is the pointwise triangle in the proof of Lemma 8.11(1), which itself does not require that the Val bound hold at every valuation over a base.

For $W \in \mathcal{W}_d(X)$, the cases are as follows. If p is invertible, the right term is zero, hence $C_p(q)(W) \in D^{\geq q+1}$. If W is mixed characteristic, proposition 4.3 and (14) put the right term in $D^{\geq q}$ and hence $C_p(q)(W) \in D^{\geq q}$. If $p = 0$ in W , it is Cartier smooth and

$$C_p(q)(W) \simeq \Omega_{W, \log}^q[-(q-1)].$$

This is (Bachmann et al., 2025, arXiv v1, Example 5.9 and proof of Lemma 8.11(2)), or follows by taking the cofiber of $F_p(q)(W) = \Omega_{W, \log}^q[-q]$ to its zero generic-fibre étale term. All cases give $D^{\geq q-1}$. Proposition 4.6(1) proves the bound on Y .

For injectivity, let c in degree $q-1$ restrict to zero at every characteristic- p residue field. The bound just proved allows use of proposition 4.6(2). At a trdeg-0 point W where p is a unit or W is mixed, its degree $q-1$ group is already zero. If $p = 0$ in W , its restriction to $\text{Frac } W$ factors through the residue field of the image of its generic point, where the given class is zero. The map $\Omega_{W, \log}^q \rightarrow \Omega_{\text{Frac } W, \log}^q$ is injective. Precisely this unconditional valuation injectivity is applied in (Bachmann et al., 2025, arXiv v1, proof of Lemma 8.11(3), equation (8.12)); it follows from the torsionfreeness of valuation differential forms. Thus c dies at every trdeg-0 point, and bounded detection kills it. $\square$

Lemma 4.8 (A residue-preserving valuation). *Let R be regular local of dimension $e \geq 1$ with $0 \neq p \in \mathfrak{m}_R$. There is a valuation ring C of $\text{Frac } R$ dominating R with $\kappa(C) = \kappa(R)$. Its henselization $V = C^h$ is mixed characteristic, induces the same residue-field isomorphism, and has fraction field algebraic over $\text{Frac } R$.*

Proof. Choose a regular system of parameters $(x_1, \dots, x_e)$ and put $R_i = R/(x_1, \dots, x_i)$. For $1 \leq i \leq e-1$, the localization of R_{i-1} at (x_i) is a DVR O_i with residue field $\text{Frac } R_i$. Begin with the DVR $C_e = R_{e-1}$, whose residue field is $\kappa(R)$. Descending on i , put

$$C_i = \{z \in O_i : z \bmod \mathfrak{m}_{O_i} \in C_{i+1}\}.$$

This is the composite valuation ring of O_i and the valuation ring C_{i+1} in its residue field (The Stacks Project Authors, 2026, Tag 088Z). It is a valuation ring of $\text{Frac } R_{i-1}$ and has the residue field of C_{i+1} . The image of R_{i-1} in $\text{Frac } R_i$ lies in $R_i \subset C_{i+1}$. Thus C_i contains R_{i-1} , and the map to its residue field is the residue map of R_{i-1} . It follows that $C = C_1$ dominates R and preserves its residue field. Henselization preserves the residue field and the valuation-ring property (The Stacks Project Authors, 2026, Tags 04GN and 0ASK); ind-etaleanness gives the fraction-field algebraicity. Since R is a domain and $p \neq 0$ in R , its fraction field and that of V have characteristic zero, and p remains in the maximal ideal. $\square$

Proposition 4.9 (The cdh Beilinson–Lichtenbaum bound). *Let Y be regular excellent noetherian, of dimension at most three, and suppose that p is a nonzerodivisor on Y . The dimension-three proof invokes Cossart–Piltant’s resolution theorem. Then, for every $q \geq 0$,*

$$\text{fib}(F_p(q)(Y) \longrightarrow \mathcal{E}_p(q)(Y)) \in D^{\geq q+1}. \quad (31)$$

Equivalently the comparison is an isomorphism in degrees at most $q-1$ and injective in degree q . All primes, including 2 and 3, arbitrary residue fields, and arbitrary ramification are allowed. For dimension at most two, the dimension-three resolution theorem is unnecessary.

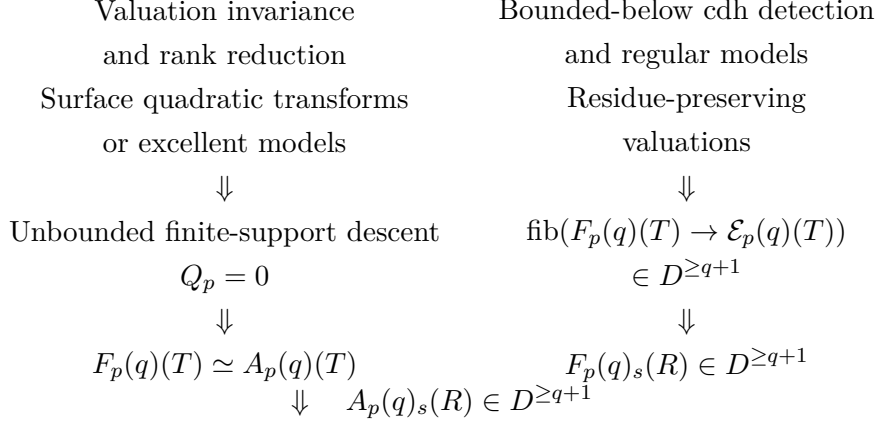
Proof. First assume Y separated and take $X = Y$ in lemma 4.7. We know that $C_p(q)(Y) \in D^{\geq q-1}$ and that (29) is injective. Let $c \in H^{q-1}C_p(q)(Y)$, and let x have characteristic- p residue field. The ring $R = \mathcal{O}_{Y,x}$ is a regular local domain of positive dimension, with $0 \neq p$ in its maximal ideal. Use lemma 4.8 to obtain $g : \text{Spec } V \rightarrow Y$ whose closed point is x and whose residue field is equal to $\kappa(x)$. Its fraction field is algebraic over the function field of the component of Y containing x ; hence $V \in \mathcal{W}_{\dim Y}(Y)$. Proposition 4.3 makes it F -smooth, with Cossart–Piltant’s resolution theorem used only when the dimension is three. The mixed-characteristic case of (30) gives $C_p(q)(V) \in D^{\geq q}$, so $g^*c = 0$. Restricting further to $\kappa(V) = \kappa(x)$ shows that $c|_{\kappa(x)} = 0$. This holds at every characteristic- p point; injectivity of (29) yields $c = 0$. Therefore $C_p(q)(Y) \in D^{\geq q}$. Shifting by $[-1]$ gives (31).

For nonseparated Y , take a finite affine covering. Every multiple intersection is an open subscheme of an affine scheme and thus is separated, regular, excellent, and satisfies the same dimension and p -torsionfreeness bounds. The comparison fibre satisfies Zariski descent. Since $D^{\geq q+1}$ is closed under limits, its bound on the Čech terms gives the bound on Y . $\square$

The equality of residue fields in lemma 4.8 is essential: in $K = \mathbf{F}_p(t)$, the logarithmic differential $d \log t = dt/t$ is nonzero, but its image in $L = \mathbf{F}_p(u)$ with $t = u^p$ is $d \log(u^p) = 0$. The nonzerodivisor hypothesis is also essential to this bound; in characteristic p the logarithmic term lies one degree lower. The proof uses only $\mathcal{W}_d(Y)$, rather than the unrestricted valuation hypothesis of (Bachmann et al., 2025, arXiv v1, Section 8.1). The proper part of descent is handled by abstract blow-up squares. Only the Nisnevich part uses a Čech nerve, whose terms preserve the dimension bound.

5 Integral purity and regular pairs

The mixed-characteristic cases of Theorem A use the following proof dependencies. The two branches are independent:



Here $T = R, U$, where $U = \mathrm{Spec} R \setminus \{s\}$, and $\mathcal{E}_p(q)_s(R) = 0$. In the nonexcellent surface case, the right-hand bound follows from the comparison $M_p \simeq F_p$ in the first assertion of [theorem 4.2](#), together with the F-smooth Beilinson–Lichtenbaum bound for M_p . In the excellent case, the independent right branch uses the cofiber $C_p(q)$, whose uniform global bound is only $D^{\geq -1}$. On the selected bounded-dimensional points it first produces $D^{\geq q-1}$ and the residue restriction injection; on mixed-characteristic regular models the cofiber bound is $D^{\geq q}$, not $D^{\geq q+1}$. The latter bound belongs to the fiber $C_p(q)[-1]$. Neither branch uses the other’s conclusion.

Proposition 5.1 (The supported residue-characteristic bound). *Let R be regular noetherian local, with closed point s and $0 \neq p \in \mathfrak{m}_R$. Suppose $\dim R \leq 2$, or that $\dim R \leq 3$ and R is excellent. Then*

$$A_{p^m}(q)_s(R) \in D^{\geq q+1} \quad (q \geq 0, m \geq 1).$$

Proof. Dimensions at most two. First replace R by R^h . The finite-coefficient theories A_{p^m} are finitary Nisnevich sheaves ([Bachmann et al., 2025](#), arXiv v1, Theorem 1.1(3)). Supported Nisnevich excision and finitariness identify the supported values of M_p , F_p , and A_{p^m} on the two rings. Henselization preserves regularity, noetherianity, dimension, and the residue field.

Write $U = \mathrm{Spec} R^h \setminus \{s\}$. The proof of [theorem 4.2](#) gives $\mathcal{N}_p(q)(U) = Q_p(q)(U) = 0$, since $\dim U \leq 1$. Together with that theorem on R^h , this identifies

$$M_p(q)_s(R^h) \simeq F_p(q)_s(R^h) \simeq A_p(q)_s(R^h).$$

Both $\mathrm{Spec} R^h$ and U are regular and p -torsionfree, hence F-smooth. By ([Bouis, 2024](#), arXiv v3, Corollary 5.12), the complex

$$\mathcal{B}(q)(T) = \mathrm{fib}(M_p(q)(T) \rightarrow \mathcal{E}_p(q)(T))$$

lies in $D^{\geq q+1}$ for $T = \mathrm{Spec} R^h$ and $T = U$. This source bound requires p -torsionfreeness and F-smoothness, and imposes no excellence assumption. Moreover $\mathcal{E}_p(q)_s(R^h) = 0$, since $R^h[1/p] \subset U$ and $U[1/p] = \mathrm{Spec} R^h[1/p]$. Thus

$$A_p(q)_s(R^h) \simeq M_p(q)_s(R^h) \simeq \mathrm{fib}(\mathcal{B}(q)(R^h) \rightarrow \mathcal{B}(q)(U)) \in D^{\geq q+1}.$$

This proves the case $m = 1$. For $m > 1$, the exact coefficient sequences give fibre sequences

$$A_{p^{m-1}}(q)_s(R^h) \longrightarrow A_{p^m}(q)_s(R^h) \longrightarrow A_p(q)_s(R^h).$$

Since $D^{\geq q+1}$ is closed under extensions, induction on m proves the bound. Supported excision returns it to the original ring.

The excellent dimension-three case. Put $X = \operatorname{Spec} R$ and $U = X \setminus \{s\}$. The theorem [theorem 4.4](#) gives $Q_p(q)(X) = Q_p(q)(U) = 0$. Thus, for $T = X, U$,

$$\operatorname{fib}(A_p(q)(T) \rightarrow \mathcal{E}_p(q)(T)) \simeq \operatorname{fib}(F_p(q)(T) \rightarrow \mathcal{E}_p(q)(T)) \in D^{\geq q+1}$$

by [proposition 4.9](#). The bound on X uses the dimension-three resolution theorem; the bound on U uses its dimension-two clause. The two generic fibres coincide, so $\mathcal{E}_p(q)_s(R) = 0$. Taking the fiber of these two comparison fibers gives $A_p(q)_s(R) \in D^{\geq q+1}$. The coefficient sequences then give the same bound for every p^m . $\square$

For a field k of characteristic p ,

$$A_{p^m}(j)(k) = W_m \Omega_{k, \log}^j[-j] \quad (j \geq 0) \quad (32)$$

by ([Bachmann et al., 2025](#), arXiv v1, Theorem 1.1(5)). Hence the closed-point purity source in dimension d is concentrated in degree $q + d$, or is zero when $q < d$.

Lemma 5.2 (Purity at the residue prime). *Under the hypotheses of [proposition 5.1](#), the point-purity map modulo p^m is an equivalence in every weight and for every $m \geq 1$.*

Proof. If $1 \leq d \leq 2$, the supported bound and the source calculation give

$$\Delta_{p^m}(q) \in D^{\geq q+d-1}.$$

In particular the cofiber lower bound is q in dimension one. Start with $\Psi_{p^m}^0 = 0$ from [lemma 2.2](#). If $\Psi_{p^m}^q = 0$, then

$$\Delta_{p^m}(q)[2q-1] = \operatorname{gr}^q \Psi_{p^m} \simeq \Sigma \Psi_{p^m}^{q+1}$$

is $(q-d+1)$ -connective. Thus $H^n \Delta_{p^m}(q) = 0$ for $n \geq q+d-1$. The interval $[q+d-1, q+d-2]$ is empty. Induction gives $\Delta_{p^m}(q) = \Psi_{p^m}^{q+1} = 0$ in every nonnegative weight.

Let $d = 3$. The supported bound gives $\Delta_{p^m}(q) \in D^{\geq q+1}$. By (2), values on R and its puncture have upper bounds $q+3$ and $q+2$, respectively. The source has degree $q+3$. Consequently

$$\Delta_{p^m}(q) \in D^{[q+1, q+3]}.$$

Put $B_q = \Delta(q)[-1]$. The same integral cofiber underlies all reductions, so

$$B_q/p^m \in D^{[q+2, q+4]}, \quad \operatorname{gr}^q \Psi_{p^m} = (B_q/p^m)[2q], \quad \Psi_{p^m}^0 = 0.$$

Tail connectivity tends to infinity by [lemma 2.2](#), and the exact eigenweights of [proposition 2.5](#) give $(k^r - 1)d_r = 0$ for every k prime to p . Apply [lemma 2.6](#) with $\ell = p$, $\epsilon = 0$ and numerical window parameter $N = 4$. This parameter is the shifted cohomological bound, while the local ring has dimension three. Since $4 \leq p+2$ for every prime, $B_q/p = 0$. The lemma uses only the reductions modulo p and p^2 at the endpoint, and its universal-coefficient argument includes $p = 2$. Coefficient extensions give $\Delta_{p^m}(q) = 0$ for every m . Negative weights vanish in both cases. $\square$

Lemma 5.3 (Integral purity on values). *For a regular noetherian local ring R with fraction field of characteristic zero, if $\dim R \leq 2$, or if $\dim R \leq 3$ and R is excellent, then*

$$A(q-d)(\kappa)[-2d] \xrightarrow{\sim} A(q)_s(R) \quad (q \in \mathbf{Z}).$$

Proof. The dimension-zero case is the identity. Let $d \geq 1$. The integral cofiber $\Delta(q)$ has zero rationalization by [proposition 2.7](#). For every prime different from the residue characteristic, Theorem [B](#) on values gives $\Delta(q)/\ell = 0$. If the residue characteristic is $p > 0$, then $p \neq 0$ in R and [lemma 5.2](#) gives $\Delta(q)/p = 0$. In residue characteristic zero every prime is invertible. Exactness of purity identifies these reductions with the cofibers of the corresponding coefficient maps. Thus multiplication by every prime is invertible on every $H^n \Delta(q)$. These groups are uniquely divisible and equal to their rationalizations, which are zero. Therefore $\Delta(q) = 0$. $\square$

5.1 Detection and smooth lifts

Lemma 5.4 (Supported continuity and the purity transformation). *Let X be noetherian of finite Krull dimension, let $i : Z \hookrightarrow X$ be a regular closed immersion, and let $f : X' \rightarrow X$ be a cofiltered limit of separated smooth finite-type maps $f_\alpha : X_\alpha \rightarrow X$ with affine dominant transition maps. Assume that X' is noetherian of finite Krull dimension. Put $Z_\alpha = Z \times_X X_\alpha$ and $Z' = Z \times_X X'$, and denote the induced maps by i_α, i' and $g : Z' \rightarrow Z$. For $E = H$ or H/n , where $H_T = s_0 KGL_T$, the canonical exceptional exchange map*

$$\chi_{i,f}(E) : g^* i^! E_X \longrightarrow i'^! E_{X'}$$

is an equivalence. With $N_{i'} \simeq g^ N_i$, the following square commutes:*

$$\begin{array}{ccc} g^*(\mathrm{Th}(-N_i) \otimes E_Z) & \xrightarrow{g^* \mathrm{pur}_i(E_X)} & g^* i^! E_X \\ \downarrow \simeq & & \downarrow \chi_{i,f}(E) \\ \mathrm{Th}(-N_{i'}) \otimes E_{Z'} & \xrightarrow{\mathrm{pur}_{i'}(E_{X'})} & i'^! E_{X'}. \end{array} \quad (\mathrm{SC})$$

Here the purity maps are the fundamental-class transformations, with the induced orientation for H/n . An equivalence of Cartesian coefficient objects transports this statement by naturality, as in [Section 1](#).

More explicitly, put $A_E(q) = A(q)$ for $E = H$ and $A_E(q) = A_n(q)$ for $E = H/n$. For a smooth test $V \rightarrow X'$ descending to compatible smooth tests $V_\alpha \rightarrow X_\alpha$, put $Z_{V_\alpha} = V_\alpha \times_X Z$ and $Z_V = V \times_X Z$. Then

$$\mathrm{colim}_\alpha A_E(q)_{Z_{V_\alpha}}(V_\alpha) \xrightarrow{\sim} A_E(q)_{Z_V}(V). \quad (\mathrm{SV})$$

The exchange in (SC) induces this canonical comparison on supported values. The conventions are $T^q E = E(q)[2q]$ and $\mathrm{map}(\mathbf{1}_V, E_V(q)) = A_E(q)(V)$. Thus (SV) has no additional $[2q]$.

Proof. We use three precise inputs. The absolute slice coefficient is Cartesian by ([Bachmann et al., 2025](#), arXiv version 1, Theorem 9.3(2)); its values and their shifts are ([Bachmann et al., 2025](#), Definition 4.15 and Remark 4.21). The presheaves $A(q)$ are finitary Nisnevich sheaves by ([Bachmann et al., 2025](#), Theorem 1.1(3)); negative weights are zero by its Remark 4.20. Finally, continuous evaluation on smooth tests, for every $F \in \mathbf{SH}(X)$, is

$$\mathrm{colim}_\alpha \mathrm{map}_{X_\alpha}(\Sigma^\infty(V_\alpha)_+, f_\alpha^* F) \xrightarrow{\sim} \mathrm{map}_{X'}(\Sigma^\infty V_+, f^* F). \quad (\mathrm{CE})$$

This is (Hoyois, 2015, arXiv version 5, Definition A.1 and Lemma A.7(1)): varying the suspension index in that lemma proves the assertion on every homotopy group, hence on mapping spectra. Its affine and dominance hypotheses are included in the statement above. Smooth finite-presentation tests descend, so these tests and their suspensions include a generating family.

Write $\Gamma_Z E_X = i_* i^! E_X$. Closed base change identifies

$$f^* \Gamma_Z E_X \simeq i'_* g^* i^! E_X.$$

Under this identification $i'_*(\chi_{i,f}(E))$ is the canonical map

$$f^* \Gamma_Z E_X \longrightarrow \Gamma_{Z'} E_{X'} \quad (\text{EX})$$

induced by the adjunction counit and the Cartesian identification of coefficients. These are proper base change and exceptional exchange from (Déglise et al., 2021b, arXiv version 3, Paragraphs 2.1.2(2),(5) and 2.1.9).

Apply (CE) to $F = \Gamma_Z E_X(q)$. At every finite stage f_α is smooth, so smooth exceptional base change identifies $f_\alpha^* \Gamma_Z E_X$ with $\Gamma_{Z_\alpha} E_{X_\alpha}$. Localization and smooth base change along $V_\alpha \rightarrow X_\alpha$ then identify its evaluation with

$$\text{fib}(A_E(q)(V_\alpha) \longrightarrow A_E(q)(V_\alpha \setminus Z_{V_\alpha})).$$

The same calculation identifies the evaluation of the target of (EX) on V with $A_E(q)_{Z_V}(V)$. The complement is quasi-compact, and its inverse system has affine transition maps and limit $V \setminus Z_V$. Finitarity and exactness of filtered colimits therefore give (SV). Reduction modulo n , being a cofiber, commutes with these fibers and filtered colimits. By the adjunction definition of exchange, the evaluation of (EX) is exactly this map of restriction fibers, rather than an unspecified isomorphism between their endpoints. Thus (EX) is an equivalence on all smooth suspension generators and all twists. It is an equivalence in $\mathbf{SH}(X')$; the fully faithful functor i'_* reflects equivalences, proving the assertion about χ .

It remains to identify the map, not just its two objects. The map f is flat, since it is a limit of smooth maps. Therefore the base-change square of i is Tor-independent and its normal bundle is the pulled-back normal bundle. The zero-excess case of (Déglise et al., 2021b, arXiv version 3, Proposition 3.2.8), applied to this square, identifies the pulled-back fundamental class with $\eta_{i'}$. This proposition allows an arbitrary base-change map; the closed immersions themselves are of finite presentation here. Tensoring that class with the coefficient, followed by the exchange map of Paragraph 2.1.10, gives precisely the transformation of Paragraph 2.5.1. Its transverse base-change square is Proposition 2.5.4(ii)(a), namely (SC). This proves the required compatibility also at the limit. We have proved invertibility of exchange for the named coefficients under the stated limit hypotheses; no exceptional-exchange assertion for arbitrary coefficients under arbitrary ind-smooth maps has been used. $\square$

Remark 5.5 (The local and henselian systems used here). An ordinary local scheme and its henselization are limits of affine Zariski and pointed affine *étale* neighborhoods, respectively. In each neighborhood discard the irreducible components not containing its marked point, then take an affine open neighborhood of that point. These choices are cofinal. Every transition is open and its image contains the marked point of the target; it therefore meets every irreducible component of the target and is dominant. Morphisms between these affine schemes are affine. This verifies the specific hypotheses of (CE) for the local and henselian systems, without excellence or a residue-field perfectness hypothesis. Neighborhoods at a point of a smooth test are again smooth over the original base.

Consequently (CE), which is coefficient-independent, gives the stalk identifications in lemma 5.6. For $v \in V$ over $t \in X$, the ordinary local map and the induced map of henselizations factor the pullback of an arbitrary object to $\mathrm{Spec} \mathcal{O}_{V,v}$ or $\mathrm{Spec} \mathcal{O}_{V,v}^h$ through the corresponding local scheme of X . Vanishing there makes the relevant stalk zero. For the Nisnevich assertion one applies this to all *étale* tests over each smooth V , which are themselves smooth over X . The needed finite-dimensional descent bounds are (Clausen and Mathew, 2021, arXiv version 3, Theorems 3.2, 3.12 and 3.18): Theorem 3.12 is the Zariski bound and Theorem 3.18 the Nisnevich bound. The local-detection argument and then the costalk induction in lemma 5.6 therefore apply as stated.

Lemma 5.6 (Local and henselian point-costalk detection). *For a noetherian scheme T of finite Krull dimension, an object $E \in \mathbf{SH}(T)$ is zero if its restriction to $\mathrm{Spec} \mathcal{O}_{T,t}$ is zero for every $t \in T$. The same conclusion holds if its restrictions to $\mathrm{Spec} \mathcal{O}_{T,t}^h$ are zero for every t .*

Let T be noetherian of finite Krull dimension and $E \in \mathbf{SH}(T)$. For $h_t : T_t = \mathrm{Spec} \mathcal{O}_{T,t}^h \rightarrow T$ and its closed point a_t , if $a_{t} h_t^* E = 0$ for every t , then $E = 0$.*

Proof. The coefficient-independent mapping-continuity formula (CE) and the local systems in remark 5.5 apply here. For V smooth over T and $v \in V$ over t , the Zariski stalk of the spectrum-valued sheaf $W \mapsto E(W)(q)$ at v is a continuous colimit of values on tests over $\mathrm{Spec} \mathcal{O}_{T,t}$, hence zero. A Zariski sheaf of spectra on the noetherian space V of finite dimension is hypercomplete, so $E(V)(q) = 0$ for all V, q . The smooth suspension spectra and their Tate shifts generate $\mathbf{SH}(T)$, so $E = 0$. The finite-dimensional hyperdescent bound is the Zariski case of (Clausen and Mathew, 2021, arXiv version 3, Theorems 3.2, 3.12 and 3.18).

For the henselian version, fix $v \in V$ over t . Henselization commutes with this local base change: $\mathcal{O}_{V,v}^h$ is the henselization at the corresponding point of $V \times_T \mathrm{Spec} \mathcal{O}_{T,t}^h$ (The Stacks Project Authors, 2026, Tag 08HU). That fibre product is smooth over the henselian local base. Its local henselization is a continuous limit of *étale* neighbourhoods of such smooth tests. Thus the vanishing of the base-restricted object makes the Nisnevich stalk $E(\mathrm{Spec} \mathcal{O}_{V,v}^h)(q)$ zero. Finite-dimensional Nisnevich hyperdescent, with the same cited bounds, gives $E(V)(q) = 0$. The generators again conclude $E = 0$.

Costalks. We prove by induction on n that $h_t^* E = 0$ for every t with $\dim \mathcal{O}_{T,t} \leq n$. Put $F_t = h_t^* E$ and $U_t = T_t \setminus \{t\}$. For $z \in U_t$, its image $u \in T$ is a proper generization of t , since the closed fibre of henselization is exactly $\mathrm{Spec} \kappa(t)$ (The Stacks Project Authors, 2026, Tag 07QM). Thus $\dim \mathcal{O}_{T,u} < \dim \mathcal{O}_{T,t}$. The local map $\mathcal{O}_{T,u} \rightarrow \mathcal{O}_{T,z}$ induces the factorization

$$\mathrm{Spec} \mathcal{O}_{T,z}^h \xrightarrow{b_z} \mathrm{Spec} \mathcal{O}_{T,u}^h \xrightarrow{h_u} T$$

by (The Stacks Project Authors, 2026, Tag 04GS). Consequently

$$F_t|_{\mathrm{Spec} \mathcal{O}_{T,z}^h} \simeq b_z^* h_u^* E = 0$$

by induction. Henselian-local detection on the noetherian puncture gives $F_t|_{U_t} = 0$. Localization and the original closed-costalk hypothesis now give $F_t \simeq a_{t*} a_t^! F_t = 0$. For local dimension zero the puncture is empty, which starts the induction. Finite dimension makes it well founded; henselian-local detection on T concludes. $\square$

Lemma 5.7 (Local injectivity over an infinite field). *Let F be an infinite field, perfect or not, $E \in \mathbf{SH}(F)$, $X \in \mathrm{Sm}_F$ irreducible with function field $F(X)$, and $x \in X$ any point. Then for all $n, q \in \mathbf{Z}$ the restriction*

$$\pi_n E(\mathrm{Spec} \mathcal{O}_{X,x})(q) \rightarrow \pi_n E(\mathrm{Spec} F(X))(q)$$

is injective.

Proof. The input is (Colliot-Thélène et al., 1997, Theorem 3.1.1). Let W be a smooth affine irreducible F -scheme of dimension $d \geq 1$ over the infinite field F , $t_1, \dots, t_r \in W$ any points, and $Z \subset W$ closed of positive codimension. Then there are $\varphi = (\psi, v): W \rightarrow \mathbf{A}^{d-1} \times \mathbf{A}^1$, an open $V \subset \mathbf{A}^{d-1}$, and an open $U \subset \psi^{-1}(V)$ containing the t_i , such that: (i) $Z \cap U = Z \cap \psi^{-1}(V)$; (ii) $\psi|_Z$ is finite; (iii) $\varphi|_U$ is étale and induces a closed immersion $Z \cap U \rightarrow \mathbf{A}_V^1$; (iv) $\varphi^{-1}(\varphi(Z \cap U)) \cap U = Z \cap U$. No perfectness is assumed. Arbitrary points reduce to closed points (Colliot-Thélène et al., 1997, Lemma 3.2.1), and closed points with residue field inseparable over F are covered: the construction makes φ étale and radicial on the relevant finite set, which forces trivial residue extensions (Colliot-Thélène et al., 1997, Theorem 3.2.2(2),(3), Lemma 3.4.3, and Lemma 3.5.1).

If $\dim X = 0$, then $\mathcal{O}_{X,x} = F(X)$. Let $d = \dim X \geq 1$.

Step 0 (reduction). Let $\alpha \in \pi_n E(\operatorname{Spec} \mathcal{O}_{X,x})(q)$ map to zero in $\pi_n E(\operatorname{Spec} F(X))(q)$. Filtered colimits are exact, so α is the image of some $\alpha_W \in \pi_n E(W)(q)$, with W an affine open neighbourhood of x , and α_W restricts to zero on a dense open $W \setminus Z$, with $Z \subset W$ closed of positive codimension. By the exact sequence $\pi_n E_Z(W)(q) \rightarrow \pi_n E(W)(q) \rightarrow \pi_n E(W \setminus Z)(q)$, $\alpha_W = u(\beta)$ for some $\beta \in \pi_n E_Z(W)(q)$, where u forgets supports. If $x \notin Z$, then $W \setminus Z$ is a neighbourhood of x and $\alpha = 0$. Assume $x \in Z$.

Step 1 (presentation). Apply the presentation theorem to W , Z and $t_1 = x$. Put $Z_U = Z \cap U$ and $T = \varphi(Z_U)$, closed in $\mathbf{A}_V^1$ by (iii). By (iii) and (iv) the square with rows $U \setminus Z_U \rightarrow U$ and $\mathbf{A}_V^1 \setminus T \rightarrow \mathbf{A}_V^1$ and vertical maps induced by φ is a distinguished Nisnevich square, so $\varphi^*: E_T(\mathbf{A}_V^1)(q) \rightarrow E_{Z_U}(U)(q)$ is an equivalence by Nisnevich descent. T is finite over V : $Z_U = Z \cap \psi^{-1}(V)$ is finite over V by (i) and (ii), and T is its isomorphic image.

Step 2 (localization on V). Let $s = \psi(x) \in V$ and $V_s = \operatorname{Spec} \mathcal{O}_{V,s}$, an essentially smooth local F -scheme whose residue field $\kappa(s)$ contains F . Base change along $V_s \rightarrow V$ gives $U_s \rightarrow \mathbf{A}_{V_s}^1$ étale, T_s finite over V_s , and $Z_s = Z_U \times_V V_s$ mapping isomorphically onto T_s , with $\varphi_s^{-1}(T_s) = Z_s$. The square is a cofiltered limit of distinguished squares with affine transition maps, so by continuity again $\varphi_s^*: E_{T_s}(\mathbf{A}_{V_s}^1)(q) \simeq E_{Z_s}(U_s)(q)$.

Step 3 (a section missing T_s). The closed fibre of T_s over s is a finite subset of $\mathbf{A}_{\kappa(s)}^1$. As F is infinite, choose $c \in F$ such that the $\kappa(s)$ -rational point $t = c$ does not lie in it. The section $\sigma_c: V_s \rightarrow \mathbf{A}_{V_s}^1$, $t = c$, has $\sigma_c^{-1}(T_s)$ closed in the local scheme V_s and not containing its closed point, hence empty. By naturality of supports, $\sigma_c^* \circ u: E_{T_s}(\mathbf{A}_{V_s}^1)(q) \rightarrow E(V_s)(q)$ factors through $E_\emptyset(V_s)(q) = 0$, and σ_c^* on $E(\mathbf{A}_{V_s}^1)(q)$ is an equivalence, by $\mathbf{A}^1$ -invariance over the essentially smooth V_s and continuity. Hence $u: E_{T_s}(\mathbf{A}_{V_s}^1)(q) \rightarrow E(\mathbf{A}_{V_s}^1)(q)$ is null.

Step 4 (conclusion). φ_s^* commutes with u , so $u: E_{Z_s}(U_s)(q) \rightarrow E(U_s)(q)$ is null. The local scheme $\operatorname{Spec} \mathcal{O}_{X,x}$ maps to U_s , since $x \in U$ maps to s and a map from a local scheme sending the closed point to s factors through V_s . By naturality, α is the image of β under $E_Z(W) \rightarrow E_{Z_U}(U) \rightarrow E_{Z_s}(U_s) \xrightarrow{u} E(U_s) \rightarrow E(\operatorname{Spec} \mathcal{O}_{X,x})$, which is zero. $\square$

Proposition 5.8 (Detection by smooth function fields). *Let F be a field and $E \in \mathbf{SH}(F)$. If $E(\operatorname{Spec} L)(q) = 0$ for every $L = F(X)$ with $X \in \operatorname{Sm}_F$ irreducible and every $q \in \mathbf{Z}$, then $E = 0$.*

Proof. Let F be infinite. By lemma 5.7, $\pi_n E(\operatorname{Spec} \mathcal{O}_{X,x})(q) = 0$ for every $X \in \operatorname{Sm}_F$, $x \in X$ and $n, q \in \mathbf{Z}$, since the connected components of X are irreducible. These are the stalks of the homotopy presheaves of the Zariski sheaf of spectra $U \mapsto E(U)(q)$ on the noetherian space X of finite Krull dimension. That sheaf is hypercomplete, so $E(X)(q) = 0$, as in the proof of lemma 5.6. The $\Sigma^{a,b} \Sigma^\infty X_+$ generate $\mathbf{SH}(F)$, so $E = 0$.

Let F be finite, hence perfect. Every finitely generated extension of F is then separably generated, hence of the form $F(X)$ with X smooth. The value of E at such a field is the value there of its bigraded homotopy sheaves, since Nisnevich sheafification does not change values at fields. These sheaves are strictly $\mathbb{A}^1$ -invariant, and a morphism of strictly $\mathbb{A}^1$ -invariant sheaves over a perfect field is an isomorphism if it is so on all finitely generated fields (Hoyois, 2015, arXiv version 5, Theorem 2.7). So all homotopy sheaves of E vanish, and $E = 0$ by (Hoyois, 2015, arXiv version 5, Theorem 2.3 and Corollary 2.4). $\square$

Lemma 5.9 (Smooth lifts and a fieldwise criterion for point purity). *Let $\mathrm{Spec} B \rightarrow \mathrm{Spec} A$ be a closed immersion of affine schemes. Smooth B -schemes have a Zariski basis of open subschemes lifting to smooth A -schemes, stable under further restriction. If R is regular noetherian local of dimension d , and U is an irreducible smooth $\kappa(R)$ -scheme with such a lift $\tilde{U}$, the henselization R_L of $\mathcal{O}_{\tilde{U}, \eta_U}$ is regular noetherian local of dimension d , with residue field $L = \kappa(U)$. If R is excellent, R_L is excellent. If $p \neq 0$ in R , then $p \neq 0$ in R_L .*

Let R be regular noetherian local, s its closed point and κ its residue field. For the intrinsic coefficient H , fix coefficients $\mathbf{Z}$ or $\mathbf{Z}/n$. If point purity on R_L is an equivalence on values in every weight for every smooth function field L/κ , then point purity on R is an equivalence in $\mathbf{SH}(\kappa)$.

Proof. Locally a smooth map factors as a standard étale map to affine space (The Stacks Project Authors, 2026, Tags 054L and 02GT). Write the chart as $B'[t]_h/(f)$, with B' an affine-space coordinate algebra over B , f monic and f' a unit. Lift B' to the corresponding algebra A' over A , lift the coefficients and h , and invert both the lifted h and the lifted f' ; the resulting standard étale algebra lifts the given chart. An open chart lifts by removing its closed complement in the closed fibre. The lifting construction is explicit and uses only the cited standard local structure. Put $S = \mathcal{O}_{\tilde{U}, \eta_U}$. Smoothness over the regular R gives regularity by (The Stacks Project Authors, 2026, Tag 07NF). The ring S is noetherian, $R \rightarrow S$ is flat local, and $S/\mathfrak{m}_R S = L$, since η_U is the generic point of the reduced closed fibre. In particular $\mathfrak{m}_S = \mathfrak{m}_R S$. If $t_1, \dots, t_d$ is a regular system of parameters of R , its image is a regular system of parameters of S and of S^h ; it generates the maximal ideals by the displayed equality and its henselian base change. The conormal frame, and therefore the dual normal frame, of the closed point is the pullback of this same frame over κ . Thus the flat dimension formula (The Stacks Project Authors, 2026, Tag 00ON) gives $\dim S = d$. For $R_L = S^h$, noetherianness, regularity and dimension preservation are (The Stacks Project Authors, 2026, Tags 06LJ, 06LN and 06LK).

If R is excellent, so is S by (The Stacks Project Authors, 2026, Tag 07QU). Its henselization is a G-ring by (The Stacks Project Authors, 2026, Tag 07QR). To check J-2, write S^h as a filtered colimit of étale S -algebras S_α . Every finite-type S^h -algebra B is finitely presented and descends to a finite-type S_α -algebra B_α . The latter is excellent. The map $B_\alpha \rightarrow B$ is ind-étale, and both rings are noetherian, so it is regular by (The Stacks Project Authors, 2026, Tag 07EP). At corresponding primes its local map is flat local; regularity ascends along this regular map and descends along its faithfully flat local map (The Stacks Project Authors, 2026, Tags 0H7S and 07NG). Consequently $\mathrm{Reg}(B)$ is the inverse image of $\mathrm{Reg}(B_\alpha)$ and is open. This proves J-2 for S^h . The regular local ring S^h is Cohen–Macaulay and universally catenary by (The Stacks Project Authors, 2026, Tags 00NQ and 00NM), hence excellent. Finally $R \rightarrow S \rightarrow S^h$ is flat local, using (The Stacks Project Authors, 2026, Tag 07QM) for the last map; it is faithfully flat and therefore does not kill p .

Put $E = H$ or H/n . Let $s = \operatorname{Spec} \kappa$, $d = \dim R$, and let

$$P_s = \operatorname{cofib}(\operatorname{Th}(-N_s) \otimes E_\kappa \xrightarrow{\operatorname{pur}_s(E_R)} s^! E_R).$$

Take an irreducible smooth κ -scheme U , shrink it to a chart admitting an affine smooth lift $\tilde{U} \rightarrow \operatorname{Spec} R$, and write $L = \kappa(U)$. Denote its closed-fibre immersion by $i_U : U \hookrightarrow \tilde{U}$ and its smooth projection by $p : U \rightarrow s$. At this finite smooth stage the actual transformation fits into the square

$$\begin{array}{ccc} p^*(\operatorname{Th}(-N_s) \otimes E_\kappa) & \xrightarrow{p^* \operatorname{pur}_s} & p^* s^! E_R \\ \downarrow \simeq & & \downarrow \operatorname{Ex}^* \simeq \\ \operatorname{Th}(-N_{i_U}) \otimes E_U & \xrightarrow{\operatorname{pur}_{i_U}} & i_U^! E_{\tilde{U}}. \end{array} \quad (\text{F1})$$

This is transverse base change for the fundamental class and smooth exceptional base change, as in (SC). In particular it is a square of the given purity morphisms.

Choose a regular system of parameters of R . The induced frame of N_s pulls back to N_{i_U} and to the normal bundles in all the following restrictions. After twisting by (q) and mapping from the unit, the bottom arrow of (F1) becomes

$$A_E(q-d)(U)[-2d] \longrightarrow A_E(q)_U(\tilde{U}). \quad (\text{F2})$$

Its cofiber is $P_s(U)(q)$ under the convention that (q) is the Tate twist alone. Indeed $E(q) = T^q E[-2q]$, so mapping into $E(q-d)[-2d]$ gives exactly the left side of (F2).

Let $\eta = \eta_U$ and $S = \mathcal{O}_{\tilde{U}, \eta}$. Take the filtered system of affine principal open neighborhoods W of η in $\tilde{U}$, after restricting to the component containing η ; regularity makes this component open and closed. Their closed fibres $U_W = U \cap W$ are cofinal in the dense opens of U : any such open can be obtained by deleting its closed complement in the closed fibre and then shrinking to an affine principal neighborhood of η . Thus their limit is $\operatorname{Spec} L$, whereas $\lim W = \operatorname{Spec} S$. Their complements have limit $\operatorname{Spec} S \setminus \{\mathfrak{m}_S\}$. Apply (SC) and (SV) to these restrictions.

Next write $S^h = R_L$ as the limit of pointed affine *étale* neighborhoods of $\operatorname{Spec} S$. They can be chosen with entire closed fibre $\operatorname{Spec} L$: remove the finitely many other points of that fibre and shrink about the chosen point. At every such stage the square with open complements is a Nisnevich distinguished square. Nisnevich excision identifies its supported value with $A_E(q)_{\mathfrak{m}_S}(S)$, and transverse base change identifies the purity arrows under this excision map. Finitarity passes this same square to S^h . Write γ_W , γ_S , and γ_{R_L} for the weight- q fundamental-class purity maps, with coefficient E , for $U_W \hookrightarrow W$, $\operatorname{Spec} L \hookrightarrow \operatorname{Spec} S$, and $\operatorname{Spec} L \hookrightarrow \operatorname{Spec} R_L$, respectively. We obtain the commutative diagram

$$\begin{array}{ccc} \operatorname{colim}_W A_E(q-d)(U_W)[-2d] & \xrightarrow{\operatorname{colim}_W \gamma_W} & \operatorname{colim}_W A_E(q)_{U_W}(W) \\ \downarrow \simeq & & \downarrow \simeq \\ A_E(q-d)(L)[-2d] & \xrightarrow{\gamma_S} & A_E(q)_{\mathfrak{m}_S}(S) \\ \downarrow = & & \downarrow \simeq \\ A_E(q-d)(L)[-2d] & \xrightarrow{\gamma_{R_L}} & A_E(q)_{\mathfrak{m}_{R_L}}(R_L). \end{array} \quad (\text{F3})$$

Here $\mathfrak{m}_S = \mathfrak{m}_R S$, $\mathfrak{m}_{R_L} = \mathfrak{m}_R R_L$, and both residue fields are exactly L . The original d parameters give the pulled-back normal frame throughout; [lemma 5.9](#) gives $\dim S = \dim R_L = d$. No relative transcendence degree is added to this dimension.

Mapping spectra are exact, and filtered colimits commute with cofibers. Taking cofibers in (F3) therefore identifies $P_s(\operatorname{Spec} L)(q)$ with the cofiber of the actual closed-point purity map on R_L in weight q . The hypothesis makes all of these spectra zero. Every smooth function field has such a chart, so [proposition 5.8](#) gives $P_s = 0$. $\square$

5.2 Composition and the main purity statements

Proposition 5.10 (From point purity to regular pairs). *Let $i : Z \hookrightarrow X$ be a regular closed immersion between regular noetherian schemes of finite Krull dimension. Fix the coefficient H or H/n . If its point-purity map is invertible on every henselized local ring of X and Z , then its purity transformation along i is invertible.*

Proof. Treat constant codimension c on the open-and-closed components of Z . Write $E = H$ or H/n and use the oriented normalization

$$\gamma_i : E_Z \longrightarrow i^! E_X(c)[2c], \quad P_i = \text{cofib}(\gamma_i).$$

Fix $z \in Z$, put $x = i(z)$, and write $X' = \text{Spec } \mathcal{O}_{X,x}^h$, $Z' = \text{Spec } \mathcal{O}_{Z,z}^h$. Henselization commutes with quotients (The Stacks Project Authors, 2026, Tag 05WQ), so $Z' = Z \times_X X'$. Let h_X, h_Z be the projections and $i' : Z' \hookrightarrow X'$. The system in remark 5.5 and lemma 5.4 give a commutative diagram of the actual oriented transformations and their cofibers:

$$\begin{array}{ccccc} h_Z^* E_Z & \xrightarrow{h_Z^* \gamma_i} & h_Z^* i^! E_X(c)[2c] & \longrightarrow & h_Z^* P_i \\ \downarrow \simeq & & \downarrow \chi_{i,h_X}(E)(c)[2c] \simeq & & \downarrow \simeq \\ E_{Z'} & \xrightarrow{\gamma_{i'}} & i'^! E_{X'}(c)[2c] & \longrightarrow & P_{i'}. \end{array} \quad (\text{P1})$$

This is the specified coefficient exchange identifying the global and local defects.

Let $a : \text{Spec } \kappa(z) \hookrightarrow Z'$ be the closed point, put $b = i'a$, and write $e = \dim Z'$ and $d = \dim X' = e + c$. The conormal exact sequence, or equivalently the normal exact sequence

$$0 \longrightarrow N_a \longrightarrow N_b \longrightarrow a^* N_{i'} \longrightarrow 0,$$

identifies $[N_b] = [N_a] + [a^* N_{i'}]$. The multiplicative orientation identifies the associated Thom twists with the indicated Tate and cohomological shifts. Composition of the same fundamental classes (Déglise et al., 2021b, arXiv version 3, Proposition 2.5.4(i) and Theorem 3.3.2) is precisely the square

$$\begin{array}{ccc} E_{\kappa(z)}(-e)[-2e] & \xrightarrow{\text{pur}_a(E_{Z'})} & a^! E_{Z'} \\ \downarrow = & & \downarrow a^! \gamma_{i'} \\ E_{\kappa(z)}(-d)[-2d](c)[2c] & \xrightarrow{\text{pur}_b(E_{X'})(c)[2c]} & b^! E_{X'}(c)[2c]. \end{array} \quad (\text{P2})$$

In the right column we use the canonical composition identification $a^! i'^! \simeq b^!$. If both point-purity maps are invertible, then $a^! \gamma_{i'}$ is invertible. Since $a^!$ is exact, (P1) gives the specific costalk identification

$$a^! h_Z^* P_i \simeq \text{cofib}(a^! \gamma_{i'}) = 0. \quad (\text{P3})$$

Thus lemma 5.6 gives $P_i = 0$. The construction uses regular noetherian local rings, the indicated finite-dimensional continuity, and the two point-purity hypotheses. Excellence and residue-field perfectness are not needed for this passage from points to pairs. $\square$

Proof of Theorem A. For every smooth function field L/κ , the lifted ring R_L in lemma 5.9 has the same local dimension as R . The fraction field of R_L still has characteristic zero by faithful flatness; excellence persists when it is required. Thus lemma 5.3 applies to every R_L in every weight. The fieldwise criterion gives the intrinsic object point-purity equivalence.

For the pair statement, all henselized local rings are regular noetherian with fraction fields of characteristic zero, by flatness over $\mathbf{Z}$. They satisfy the stated dimension bound, and remain excellent in the dimension-three case. The point assertion applies to both schemes. Now proposition 5.10 proves the result. $\square$

Proof of Theorem B on objects and pairs. Every lifted R_L is regular noetherian of dimension d , and ℓ remains invertible. The value assertion already proved therefore holds on all of these rings in every weight, modulo each ℓ^m . The fieldwise criterion gives object point purity. For a regular pair, the henselized local rings of both schemes have dimension at most $\ell + 2$, and ℓ is invertible on them. Applying [proposition 5.10](#) gives the pair assertion. $\square$

6 Purity over regular DVR maps

Fundamental-class purity for Spitzweck’s coefficient is proved in the mixed-characteristic DVR setup of Frankland–Spitzweck’s Section 4 ([Frankland and Spitzweck, 2026](#), arXiv v4, Section 4, Proposition 8.3 and Corollary 8.4). The integral comparison of Section 1 and coefficient naturality identify this with purity for the intrinsic zero slice. In that setup, Corollary 8.4 gives its integral coefficient and prime-reduction module formulation; exactness further gives every finite reduction. This is a statement about the coefficient and its reductions, and does not assert purity for arbitrary modules.

For every mixed-characteristic DVR V , with residue field k and closed immersion $i_0 : \operatorname{Spec} k \hookrightarrow \operatorname{Spec} V$, Theorem A in dimension one gives the integral input

$$H_k(-1)[-2] \xrightarrow{\sim} i_0^! H_V.$$

This internal proof requires neither excellence nor perfection of k . We now extend this map along a regular ring map by explicit Popescu approximation and continuity, and combine it with perfect-field purity for Theorem C.

Lemma 6.1 (Special-fibre purity for a regular DVR algebra). *Let V be a mixed-characteristic DVR, let π be a uniformizer, and let R be a noetherian V -algebra such that $V \rightarrow R$ is a regular ring map. Put $X = \operatorname{Spec} R$, $D = \operatorname{Spec}(R/\pi)$, and let $i : D \hookrightarrow X$ be the special-fibre immersion. Then*

$$\operatorname{pur}_i(H_X) : H_D(-1)[-2] \xrightarrow{\sim} i^! H_X.$$

The assertion includes non-finite-type R and requires no excellence.

Proof. Popescu’s theorem gives a filtered presentation $R = \operatorname{colim}_\alpha R_\alpha$ with each R_α smooth over V ; see ([The Stacks Project Authors, 2026](#), Tag 07GC). At every stage smooth exceptional base change carries the DVR purity map above to purity for $D_\alpha = V(\pi) \subset X_\alpha = \operatorname{Spec} R_\alpha$. The normal line is trivialized by π . Thus, in every weight q , the supported Gysin map is an equivalence

$$\mathbf{Z}(q-1)^{\mathbb{A}^1}(D_\alpha)[-2] \xrightarrow{\sim} \mathbf{Z}(q)_{D_\alpha}^{\mathbb{A}^1}(X_\alpha).$$

Both R_α and R are flat over V . Every transition square for these divisors is consequently Tor-independent, even though the transition ring map need not be smooth: the resolution of R_α/π by multiplication by π stays exact after base change. Transverse base change for fundamental classes therefore makes these the compatible fundamental-class Gysin maps, not merely a collection of isomorphisms; use ([Déglise et al., 2021b](#), arXiv v3, Proposition 2.5.4 and Paragraph 4.3.1).

The source and the two terms defining the supported target commute with this filtered colimit. For the target these terms are the values on R_α and $R_\alpha[1/\pi]$. Filtered colimits of complexes commute with fibres. Finitarity ([Bachmann et al., 2025](#), arXiv v1, Theorem 1.1(3)) thus gives the same supported equivalence on X .

This argument also applies to every smooth affine X -scheme T . Indeed, a finite smooth presentation and its Jacobian data descend to some R_α , and its later models are smooth over V . Finite affine covers give the assertion for all smooth tests T . For these tests, mapping into the pushforward of the displayed purity transformation gives exactly

$$\mathbf{Z}(q-1)^{\mathbb{A}^1}(T \times_X D)[-2] \longrightarrow \mathbf{Z}(q)_{T \times_X D}^{\mathbb{A}^1}(T).$$

All its cohomological shifts are equivalences. Smooth suspension spectra and their Tate shifts generate $\mathbf{SH}(X)$, so $i_* \text{pur}_i(H_X)$ is invertible. Since i_* is fully faithful, $\text{pur}_i(H_X)$ is invertible. This proof uses smooth exchange at finite stages and finitary supported values; it does not assume an exceptional-exchange theorem for an arbitrary ind-smooth map. $\square$

Proof of Theorem C. The fibre $D = \text{Spec}(R/\pi)$ is regular of dimension $d-1$. The preceding lemma proves purity for $D \hookrightarrow \text{Spec } R$. Purity for $s \hookrightarrow D$ follows from (Déglise et al., 2021a, arXiv v3, Appendix C, Theorem C.1, first proof (Lemmas C.4–C.5 and Proposition C.8)), applied over $\mathbf{F}_p$. For any absolute spectrum in a continuous constructibly generated motivic category, that proof establishes integral purity on regular noetherian schemes over a perfect field, without requiring their structure maps to be finite type. Lemma C.4 and formula (C.0.a) identify this with the same intrinsic fundamental-class transformation. For $\mathbf{SH}$, continuity holds on qcqs inverse systems with affine transition maps without a dominance hypothesis (Déglise et al., 2021a, arXiv v3, Definition A.2 and Examples A.4 and A.9). The absolute coefficient H satisfies these hypotheses. Composition of the fundamental classes gives point purity on R . The normal classes add, giving the twist $(-d)[-2d]$. This proves object purity, its value assertion, and every finite coefficient assertion. Formula (32) gives the logarithmic de Rham–Witt value in the statement.

If R is unramified, it is p -torsionfree and hence flat over $\mathbf{Z}_{(p)}$. Its generic fibre is regular over the perfect field $\mathbf{Q}$. Its closed fibre R/p is regular, since p is part of a regular system of parameters, and is geometrically regular over the perfect field $\mathbf{F}_p$. Therefore $\mathbf{Z}_{(p)} \rightarrow R$ is a local regular map, as required.

For the pair assertion, the structure maps of both schemes to $\mathbf{Z}_{(p)}$ are regular on affine charts: they are flat and their generic and special fibres are regular over perfect fields. The special-fibre lemma proves purity for $j_X : X_p \hookrightarrow X$ and $j_Z : Z_p \hookrightarrow Z$. The generic-fibre immersion $i_{\mathbf{Q}}$ has regular source and target. Flatness of Z makes the square defining $i_p : Z_p \hookrightarrow X_p$ Tor-independent, so i_p is a regular immersion. Its source and target are regular, and the same integral theorem (Déglise et al., 2021a, arXiv v3, Appendix C, Theorem C.1, first proof) gives purity for $i_{\mathbf{Q}}$ over $\mathbf{Q}$ and for i_p over $\mathbf{F}_p$, in both cases for the intrinsic fundamental-class map.

The two factorizations $i \circ j_Z = j_X \circ i_p$ and composition show that $j_Z^! \text{pur}_i(H_X)$ is invertible: its composition with purity for j_Z equals the composite of purity for j_X and i_p , with the same normal classes and twists. The cone of $\text{pur}_i(H_X)$ has zero generic restriction and zero exceptional restriction to Z_p . The localization triangle on Z kills this cone. $\square$

The remaining supported-comparison boundary. For a general regular mixed-characteristic local ring, the bound $A_p(q)_s(R) \in D^{\geq q+1}$ is not supplied here beyond the low-dimensional or regular-DVR-map cases. A bound for M_p alone does not give this bound after $\mathbb{A}^1$ -localization. If such a supported bound is supplied independently in dimension $d \geq 2$, the shifted Adams argument has numerical parameter $N = d + 1$ and gives residue- p value purity for $d \leq p + 1$. In dimension one, the connectivity squeeze in lemma 5.2 gives value purity. The invertible-prime argument proves purity in the range $d \leq \ell + 2$; no optimal range for geometric purity is asserted. The nonexcellent surface argument does not supply

valuation unions in dimension three; that case here retains its excellent regular-model and resolution inputs. No general ramified higher-dimensional integral assertion is inferred from these methods.

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